



Block-2

Regulating Internet

THE PEOPLE'S
UNIVERSITY

BLOCK 2: REGULATING INTERNET

Introduction

Unit 6: ‘The Concept of Security in Cyberspace’ explores the need and necessity of computer security concerns, various security challenges of cyber security and explains the differences between computer-related cyber crimes and computer-facilitated cyber crimes.

Unit 7: ‘Cyberspace and Crime’ describe the characteristics of cyberspace from the surveillance and security perspectives and the difference between real space and cyberspace. This Unit also explores the different dimensions of digital identity, cyber crimes and their prevention mechanisms.

Unit 8: ‘Cyber Laws’ acquaints you with the law regulating cyberspace in India. It explains the concept of cyberspace and discusses the need for regulation of cyberspace. Further, it outlines various issues pertaining to cyberspace, cyber crimes and offences under the Information Technology Act.

Unit 9: ‘Information Technology (IT) Act’ explains the primary aims and objectives of India’s Information Technology Act, its amendments, and its criticism. More specifically, this Unit familiarises you with the various offences and their penalties and provisions relating to e-governance and legal recognition of electronic records.

Unit 10: ‘Understanding Data Journalism’ is a unit that deals in detail about how data science has become a matter of utmost importance in every public sphere—social, economic as well as political. It also stresses the importance of data for journalists and how to be skilled in the art of using data for building strong, credible stories.

We hope that this Block, in particular, and the Course in general, facilitated learning and understanding of the issues that revolve around the emerging digital media platform and how various institutions of national and international efforts are underway to make cyberspace more secure safe.

UNIT 6 THE CONCEPT OF SECURITY IN CYBERSPACE

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6.0 INTRODUCTION

Cyberspace should not be confused with the 'internet'. While the Internet is the interconnection between millions of computers located around the world, each of them is independently managed by persons who have chosen to adhere to common communications protocols, particularly a fundamental protocol suite known as Transmission Control Protocol or Internet Protocol (TCP/IP), which makes it practical for computers to share data even if they are far apart and have no direct line of communication, the term 'cyberspace' is often used simply to refer to objects and identities that exist largely within the computing network itself, so that a web site, for example, might be metaphorically said to 'exist in cyberspace'. According to this interpretation,

events taking place on the Internet are not happening in the countries where the participants or the servers are physically located, but ‘in cyberspace’.

When we sit in front of a computer and switch it on, something like magic happens before us; if we are correctly connected, we can bring up an environment of hypertext with a click of the mouse. It feels like that behind the screen, and there is a potentially very huge reservoir of information that is always in the making. Such a reservoir is somewhere, out there. We are certainly aware that people who generate information and where the information resides are not behind the screen or in the hard drive. However, we nevertheless take the computer as a gateway to another place where other people have done similar things. Conceptually, we tend to envision a nonphysical ‘space’ existing between here and there and believe that we can access that ‘space’ by utilising computer-based technologies. We send messages to others by email or talk to others in a chat room. We play an online interactive game as if our opponent (in the game) is right before us, though invisible. By participating in an online teleconference, we experience the presence of other conference participants. Nevertheless, where are we? Where are the others with whom we communicate? We seem to communicate in a not defined medium, and there is a sense of spatiality. Usually, we call this medium ‘cyberspace’, the ‘space’ that seems to open up or shut down as the computer screen is activated or deactivated.

6.1 LEARNING OUTCOMES

After studying this Unit, you should be able to:

- know the meaning of ‘cyberspace’;
- know the reasons for security concerns in cyberspace;
- explain the need to secure cyberspace;
- describe the specific security challenges to cyberspace;
- explain the concept of Cyber Security; and
- know the distinction between computer-related and computer-facilitated crime.

6.2 CYBERSPACE – WHY IS IT NOT SECURE?

The TCP/IP protocol suite makes the Internet possible. Its most important feature is defining a packet-switching network, a method by which data can be broken up into standardised packets that are then routed to their destinations via an indeterminate number of intermediaries. Under TCP/IP, as each intermediary receives a packet intended for a party further away, the packet is forwarded along whatever route is most convenient at the nanosecond the data arrives. So in simple terms, if you desire to deliver a package to a friend, instead of sending it as one whole piece, you send parts of the package through many people who are heading in the direction of the

recipient. If one such person carrying part of the package meets another person who is heading to meet the recipient, then this person hands over his parts to the other person; eventually, the parts finally reach the recipient. In this model, you can see some level of confusion. The package goes all over space before reaching its final destination; it also exchanges or moves hands.

Another reason for insecurity associated with cyberspace is its end-to-end design. The network is designed so that any form of intelligence, including security interfaces, is placed only at the 'ends'. You may call these interfaces firewalls, filters, spam killer, diluter, etc. With the end-to-end design, the network has minimal control or intelligence to intercept the manner of its usage. Computers within the network are only required to provide the most basic level of service — data transport via the TCP/IP protocols. The network itself is kept simple, incapable of discrimination. Without intelligence embedded in the network, all packets that conform to the protocol are transmitted, regardless of content, regardless of intent, and without any knowledge (or care) of what types of applications or people are utilising the packets on the ends of the network.

Further, the Internet is not controlled by a single company or agency. The only organisation that exerts some level of Internet monitoring is an international, unincorporated organisation called the Internet Engineering Task Force. A primary activity of the IETF is internet standard-setting. The Internet Standards Process (ISP) is concerned with all protocols, procedures, and conventions that are used in or by the Internet, including the TCP/IP protocol suite.

The technology – packet-switching protocols, the end-to-end network design and the impossibility of centralised control make computer networks or cyberspace an anarchic ethos. The endeavour is to bring sublimity to the confusion and then, over a period of time, instil control that will assume some checks and provide security to the “network”.

Check Your Progress: 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1. What are the three main reasons for contributing lack of security to cyberspace?

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6.3 WHY SHOULD WE SECURE CYBERSPACE?

Societies are becoming more dependent on computer networks and, therefore, more vulnerable to cybercrime and terrorism. Increasing traffic of commerce and trade in cyberspace is adding to its woes of being a target of

creative misuse. Creative misuse, because each day something new is created, or someone new is creating some software or intermediary, to vandalise or penetrate into cyberspace to perpetrate fraud, disrupt the transmission of information and data. Hackers find thrill in penetrating networks and destroying data, while terrorists could purposely disrupt the critical infrastructures that are dependent on networked computers. Electronic communication and transmission of data are still not secure. At the same time, consumers hesitate from disclosing personal and credit card data on the Internet, with security and privacy being their primary concerns. Businesses face loss of proprietary data, intellectual property and online access to customers and suppliers due to security breaches and intentional service interruptions. For 'cyberspace' to contribute to economic growth, human development and democratisation, it must be trustworthy and secure. Lack of trust and security jeopardises development goals that could be supported by trusted cyberspace.

Securing cyberspace is by no means an easy task. Its universal spread across all nations without the barriers of sovereign controls makes the challenge of monitoring cyberspace even more difficult. It will be impossible to secure cyberspace through law without the application of technology and participation of every individual who accesses cyberspace. The United States Government, while implementing its National Strategy to Secure Cyber, observes that cyberspace by its very nature is chaotic and beyond the reach of any organised or central control.

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1. What are the specific losses that businesses face on account of unsecured cyberspace?

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6.4 SECURITY CHALLENGES IN CYBERSPACE

As discussed earlier, cyberspace faces the constant threat of creative misuse. How does one ensure to minimise this threat? The best way to begin is by being aware of the kind of cyber threats and then using this awareness to find means of controlling the violation. This process, as you will see, identifying the threat and then adequately protecting against them is an ongoing process of legislation, technological innovation, and cooperation amongst users, on a micro-level amongst users and amongst nations at the macro user level. Cybercrimes consist of specific crimes dealing with computers and networks, such as hacking, phishing and the facilitation of traditional crime through the use of computers (child pornography, hate crimes, telemarketing/internet

fraud). In addition to cybercrime, there is also a computer-supported crime that covers the use of computers by criminals for communication and data storage. A brief introduction to some common cyber-related violations or cybercrimes as they are more commonly referred to are discussed below:

6.4.1 Hacking

Hacking in simple terms means an illegal intrusion into a computer system and/or network. There is an equivalent term to hacking, i.e. cracking, but from an Indian legal perspective, there is no difference between the term hacking and cracking. Every act committed to breaking into a computer and/ or network is hacking. Hackers write or use ready-made computer programs to attack the target computer. Some hackers hack for personal monetary gains, such as stealing credit card information or transferring money from various bank accounts to their own, followed by money withdrawal. They also indulge in extortion based on information received while hacking a particular computer system network.

6.4.2 Child Pornography

The Internet is extensively used for the sexual abuse of children. As more homes have access to the Internet, more children are accessing it, enhancing their vulnerability of falling victims to the aggression of paedophiles.

Easy access to pornographic content readily and freely over the Internet lowers children's inhibitions. Paedophiles lure children by distributing pornographic material and pursuing them for sexual exploitation. Sometimes paedophiles contact children in chat rooms posing as teenagers or children of similar age; they win the confidence of these children, then induce them into sexually provocative discussions. Then begins the actual exploitation of children.

6.4.3 Cyber Stalking

There is no universally accepted definition of cyberstalking; the term refers to the use of the Internet, email, or other electronic communications devices to stalk another person. Stalking generally involves harassing or threatening behaviour that an individual engages in repeatedly, such as following a person, appearing at a person's home or place of business, making harassing phone calls, leaving written messages or objects, or vandalising a person's property. Most stalking laws require that the perpetrator make a credible threat of violence against the victim; others include threats against the victim's immediate family. Still, others require only that the alleged stalker's conduct constitutes an implied threat.

Cyberstalking can be defined as the repeated acts of harassment or threatening behaviour of the cybercriminal towards the victim by using internet services. The modus operandi of most stalkers is as follows:

- a) Collect personal information about the victim. Suppose the stalker is a stranger to the victim. In that case, he collects the information from

internet resources such as various profiles the victim may have filled in while opening a chat or email account or signing an account with some website.

- b) The stalker may post information on any website related to sex services or dating services, posing as if the victim is posting this information and inviting the people to call the victim on her telephone to obtain sexual services.
- c) Some stalkers subscribe to the victim's email account to innumerable pornographic and sex sites, because of which the victim starts receiving indecent soliciting emails.
- d) Some stalkers send repeated emails asking for various favours or threatening the victim.

6.4.4 Denial of Service

This is a technology-driven cyber intrusion, whereby the influencer floods the bandwidth or blocks the user's mails with spam mails depriving the user, access to the Internet and the services provided from that place. A DoS Attack (as it is commonly known) can be perpetrated in a number of ways.

There are three basic types of attack:

- a) consumption of computational resources, such as bandwidth, disk space, or central processing unit (CPU) time;
- b) disruption of configuration information, such as routing information; and
- c) disruption of physical network components.

6.4.5 Dissemination of Malicious Software (Malware)

Malware is defined as software designed to perform an unwanted illegal act via the computer network. It is also loosely defined as software with malicious intent. Malware can be classified based on how they get executed, how it spread, and/or what they do. Some of them are discussed below.

a) Virus

The term "computer virus" was first termed by Fred Cohen in 1983 in his PhD thesis. A virus is a program that can infect other programs by modifying them to include a possibly evolved copy of itself. A virus can spread throughout a computer or network using every user's authorisation to infect their program. Every infected program may also act as a virus, and thus the infection grows. Viruses normally affect program files, but in some cases, they also affect data files, disrupting data use and destroying them completely.

b) Worms

Worms are also disseminated through computer networks; unlike viruses, computer worms are malicious programs that copy themselves from system to system rather than infiltrating legitimate files. For example, a mass mailing email worm is a worm that sends copies of itself via email. On the other hand, a network worm makes copies of itself throughout a network, thus disrupting an entire network.

c) Trojans

A trojan is another form of Malware, and Trojans do things other than what is expected by the user. Trojan or Trojan horse is a program that generally impairs the security of a system. The program is usually disguised as something else (a benign program) or masqueraded as a legitimate file that the user would expect to see or load on the system. The payload of a Trojan is usually delivered as soon as it is opened and usually with devastating results. Trojans are used to create back-doors (a program that allows outside access into a secure network) on computers belonging to a secure network so that a hacker can have access to the secure network. Also, Trojans are often associated with remote access programs that perform illicit operations such as password-stealing or which allow compromised machines to be used for targeted denial of service.

d) Hoax

Hoax is an email that warns the user of a certain system harming the computer. The message after that instructs the user to run a procedure (most often in the form of a download) to correct the harmful system. When this program is run, it invades the system and deletes an important file.

e) Spyware

Spyware invades a computer and, as its name implies, monitors a user's activities without consent. Spywares are usually forwarded through unsuspecting emails with bonafide email i.ds. Spyware continues to infect millions of computers globally.

6.4.6 Phishing

Phishers lure users to a phoney website, usually by sending them an authentic email. Once at the fake site, users are tricked into divulging various private information, such as passwords and account numbers.

6.4.7 Information Warfare

Information warfare is a kind of warfare where information and attacks on information and its system are used as a tool of warfare. Information warfare may include giving the enemy propaganda to convince them to give up and denying them information that might lead to their resistance. Information warfare may also include feeding propaganda or even disinformation to one's

own population, either to build support for the war effort or counter enemy propaganda.

Information warfare may also mean a strategy for undermining an enemy's data and information systems while defending and leveraging one's own information edge. This type of war has no front line; potential battlefields are anywhere networked systems can be accessed-oil and gas pipelines, electric power grids, telephone switching networks, etc.

Information warfare can take countless forms: trains and planes can be misrouted and caused to collide, stock exchanges can be sabotaged by electronic "sniffers" which disrupt international fund-transfer networks, and the signals of television and radio stations can be jammed and taken over and used for a misinformation campaign.

During the Gulf War, Dutch crackers stole information about U.S. troop movements from U.S. Defence Department computers and tried to sell it to the Iraqis, who thought it was a hoax and turned it down. In January 1999,

U.S. Air Intelligence computers were hit by a coordinated attack, part of which appeared to come from Russian cracking.

6.4.8 Data Related

Data interception - Hijacking emails, interference of an intermediary in the network, maybe a prelude to another type of computer crime, typically data modification.

Data modification - Usually done in conjunction with data interception, valid data intended for a recipient is hijacked or intercepted and then is replaced with an erroneous one. This could also apply to illegal tapping into the database and altering its contents. Any form of alteration without appropriate authorisation falls under this category.

Data theft - outright stealing, most commonly classified or proprietary information without authorisation. This could be the result of data interception. It might also be the unlawful use or possession of copyrighted works such as songs, pictures, movies or other works of art.

6.4.9 Network Related

Network interference - any activity that causes the operation of a computer network to be temporarily disrupted. Interference implies something momentarily, such as Denial of Service Attacks that cause delays in data transmission by using up all available bandwidth. Distributed denial of service, ping of death and smurf attacks also fall under this category.

Network sabotage - causing permanent damage to a computer network, such as deleting files or records from storage.

Check Your Progress: 3

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1. What are the major security challenges in cyberspace?

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2. What are the forms of data related threats in cyberspace?

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6.5 THE CONCEPT OF CYBER SECURITY

As discussed above, securing cyberspace and ensuring adequate legal and technological protection is a collaborative effort between users (including nations) and technology. Technological innovations focussed on minimising cybercrimes should be backed by substantial legislation. Hence, cyber security rests on two pillars; (i) technological innovations and end-user applications, and (ii) legislation against cybercrimes. Many are of the opinion that technology alone will be sufficient to secure cyberspace, but in the arena of social engineering, where any criminal act, or even its attempt, requires to be legally addressed for social stability, having strict legislation in place against cyber interference is essential.

6.5.1 Technology's Answers to Cyber Security

Today's technology market is booming with all types of security tools, software, hardware locks, etc. Innovation is positive. However, is technology alone sufficient to secure cyberspace without cooperation amongst its users? The answer is no. As discussed earlier, the Internet is based on end-to-end technology. So maximum control and observation may be applied at the two ends with a fair amount of control and observation in the medium. There are four types of technologies that may be applied for cyber security. When I speak of technologies, I am not referring to tools.

i) Unilateral Technology

Unilateral technologies are technologies that each user can decide upon for themselves. Therefore, neither coordination nor negotiation is needed concerning their use. Important unilateral technologies for multilateral security are:

- Tools to help even inexperienced users to formulate all their protection goals, if necessary for each and every application or even each and every action;

- (Portable) devices that are secure for their users in order to bootstrap security. The devices need minimal physical protection comprising direct input/output with their users. If they are multipurpose, an operating system providing fine-grained access control and administration of rights for applications, adhering to the principle of least privilege. This is essential to limit the spread of Trojan horses and can prevent computer viruses completely;
- Encryption of local storage media to conceal and/or authenticate its contents;
- Hiding of secret data in local multimedia contents or the local file system using steganographic techniques, to conceal not only the contents of the secret data but also its very existence;
- Watermarking or fingerprinting digital data using steganographic techniques to help prove authorship or copyright infringements; and
- You are using only software whose source code is published and well checked or the security of which is certified by a trustworthy third party having access to the complete source code and all tools used for code generation. The best technique is to combine both approaches with regard to as much of the software as possible. Only by using at least one of these two approaches can you be reasonably certain that the software you use does not contain Trojan horses. More or less, the same applies to hardware where all sources and tools used for design and production are needed as well to check for the absence of Trojan horses.

ii) Bilateral Technologies

Bilateral technologies can only be used in the communication partners cooperation. This means that some coordination and negotiation is needed concerning their use. Bilateral technologies include tools for negotiating security mechanisms and cryptographic and steganographic mechanisms for securing content.

Important bilateral technologies for multilateral security are:

- Tools to negotiate bilateral protection goals and security mechanisms.
- Cryptographic mechanisms and steganography mechanisms to secure the communication content.

iii) Trilateral Technologies

Trilateral technologies can only be used if a third party fulfils a specific task for the other participating parties. This means that more coordination and negotiation is needed concerning their use compared with unilateral – and in most cases, bilateral – technologies. Important trilateral technologies for multilateral security are:

- Tools to negotiate trilateral security mechanisms, e.g. for accountability;
- To provide an access infrastructure where users use certain access parameters while transmitting data. Users will have the liberty to change the access tools to prevent data interception;
- Security gateways provided by third parties are gaining popularity. Security gateways are provided on the Internet to facilitate limited access to participating users only. Users often use this for the exchange of confidential information at a pre-determined secured gateway; and
- Mechanisms to provide digital pseudonyms, i.e. a suitable combination of anonymity and accountability. In particular, there are mechanisms to transfer signatures between different pseudonyms of the same party securely. This is called transferring signatures between pseudonyms.

When pseudonyms are used during accountable value exchange, there are a number of possibilities for the tasks of the integrated third party:

- Identification of the user in the event of fraud (pseudonyms are certified and the certification authority knows real identities), i.e. privacy of pseudonymous parties, cannot be guaranteed.
- Mandatory deposit of payment with an active trustee to prevent fraud in spite of completely anonymous pseudonyms, i.e. privacy of the pseudonymous parties can be guaranteed.

Trilateral security technologies include public key infrastructure techniques which can use certified public keys, security gateways, and digital pseudonyms.

iv) Multilateral Technologies

Multilateral technologies can only be used if a large number of independent parties cooperate. This means that coordination and possibly negotiation are needed on a large scale. Important multilateral technologies for multilateral security are:

- Tools to negotiate multilateral protection goals and security mechanisms, e.g., anonymity and unobservability.
- Mechanisms to provide anonymity, unobservability, and unlinkability with regard to communications, i.e. protect who communicates when to whom and from where to where.

6.5.2 Cybersecurity and Law

Most countries are still unaware of the potential threats of cybercrime and are still to come up with any guidelines, let alone legislation.

As part of the legal framework promoting trust and confidence in cyberspace, every nation should have basic criminal laws against activities that attack the confidentiality, integrity, or availability of computer data and computer systems.

As seen above, there are numerous forms of cybercrimes; it may be impossible to draft legislation for each of them. However, any form of illegal cyber interference may be broadly categorised under four heads, and appropriate legislation may be drafted to address each of them. These are discussed below:

Data interception: It should be prohibited to intercept, without right, by technical means intentionally, non-public transmissions of computer data to, from or within a computer system. This crime constitutes an essential element of cyber-trust, for it protects the confidentiality of communications. For example, it makes it illegal to intercept another person's email. In some countries, interception of telephonic conversation (without prior legal sanction by way of a court order) is illegal. Similarly, the same laws may also be extended to the interception of electronic data.

Data interference: It should be a crime to intentionally damage, delete, degrade, alter or suppress data in someone else's computer without right. This provision would cover, for example, intentionally sending viruses that delete files, or hacking a computer and changing or deleting data, or hacking a website and changing its appearance. The element of intentionality is important since otherwise, producing defective software or unintentionally forwarding a virus would be a crime.

System interference: It should be a crime to intentionally cause serious hindrance without the right to the functioning of a computer system by inputting, transmitting, damaging, deleting, deteriorating, altering or suppressing computer data. This provision should cover things like Denial of Service Attacks or introducing viruses into a system in ways that interfere with its normal usage. This offence must include, as an element of the offence, the concept that there must be significant harm (e.g., a certain threshold of monetary loss) in order for an offence to occur; otherwise, ordinary online behaviour, such as sending one or just a few unsolicited emails, would be a crime, which is not sensible.

Illegal access: This is the crime of intentionally accessing others' computer systems without having rights. It can be thought of as the cyberspace equivalent of trespassing. (Looked at another way, illegal access is an offence against the confidentiality of stored data and therefore is analogous to illegal interception, which is an offence against the confidentiality of data in transit.) This crime must be carefully defined, lest it includes a common, harmless activity. In the most serious cases, the act of illegal access is part of another crime covered by the three listed above, such as data interference, or it involves another crime covered by offline law, such as theft of proprietary data (see below). In some legal systems, the definition of the crime of illegal access is limited to situations in which confidential information (medical or financial

information) is taken, copied or viewed or where there is intent to obtain confidential information or where access is obtained only by defeating security measures.

Check Your Progress: 4

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1. What are the four main broadheads for classifying illegal cyber interference?

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6.6 COMPUTER-RELATED OR COMPUTER-FACILITATED CRIME

Discussions of computer crime often extend into matters that are not crimes against computers but are crimes facilitated by the use of computers. For example, theft is a crime in every legal system, and criminal law should cover theft whether it occurs online or offline. Similarly, fraud is a crime, and ordinary fraud statutes will often use terminology that applies equally well to online conduct as it did to offline conduct. Other crimes, such as infringement of intellectual property rights or dissemination of child pornography, are not properly computer crimes—they are crimes that the use of a computer may facilitate. By carefully examining existing provisions, most computer-related crimes may be covered under existing criminal law provisions.

6.7 APPLICATION OF BASIC CRIMINAL LAW CONCEPTS

Common concepts of criminal law such as “aiding and abetting” or “attempt” may also be applied to Cybercrime. For example, launching a virus with the intent to disrupt service might be a crime under the concept of intent, even if the virus did not work as intended. Similarly, if a nation’s law has the concept of aiding and abetting, that might be applied to cyber-crime, such that one who intentionally produces a virus and provides it to another knowing or intending that it will be used to destroy data or interfere with a system may be guilty of data or network interference caused by the virus even if the virus was introduced into a network by someone else.

Let us now summarise the point covered in this Unit.

6.8 LET US SUM UP

- ‘Cyberspace’ refers to objects and identities that exist — largely within the computing network and is different from the term ‘internet’.

- There are serious threats associated with cyberspace because of the technology– packet-switching protocols, the end-to-end network design and the impossibility of centralised control.
- There is an urgent, widespread requirement to protect and secure cyberspace because it is a target of creative misuse.
- Some of the common cyber-related violations or crimes are as follows:
 - (i) Hacking
 - (ii) Child pornography
 - (iii) Cyberstalking
 - (iv) Denial of service
 - (v) Malware
 - (vi) Phishing
 - (vii) Information warfare
 - (viii) Data related violations
 - (ix) Network related violations
- Cyber Security rests upon the two pillars of (a) technological innovations and their applications by end-users and (b) legislation against cybercrimes.
- Technology's answers to cyber security include unilateral, bilateral,
- trilateral and multilateral technologies.
- Illegal cyber interference may be on account of (a) data interception, (b) data interference, (c) system interference and (d) illegal access. Common concepts of criminal law apply to cybercrime.

6.9 FURTHER READINGS

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6.10 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. The three main reasons contributing to a lack of security in cyberspace are (a) technology packet-switching protocols, (b) end to end network design and (c) impossibility of centralised control.

Check Your Progress: 2

1. The major losses that businesses face on account of insecurity in cyberspace are the loss of proprietary data, intellectual property and online access to customers, suppliers and international service interruptions.

Check Your Progress: 3

1. (a) The major security challenges in cyberspace are:
 - Hacking
 - Child pornography

- Cyber Stalking
- Denial of service
- Malware
- Phishing
- Information warfare
- Data related challenges and
- Network related challenges

2. The three major forms of data related threat in cyberspace are:

- (a) Data interception
- (b) Data modification
- (c) Data theft

Check Your Progress: 4

1. The four major heads for categorising illegal cyber interference are:

- (d) Data interception
- (e) Data interference
- (f) System interference and
- (g) Illegal access

UNIT 7 CYBERSPACE AND CYBER CRIME

Structure

- 7.0 Introduction
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7.0 INTRODUCTION

Cyberspace is a term that is not yet completely defined and has no geographical limitation. It is a term associated with the application of the Internet worldwide. It is also called a virtual space as the physical existence of cyberspace is not detectable at all. Cyberspace is “the total interconnectedness of human beings through computers and telecommunication without regard to physical geography.

For our purposes, cyberspace is a realm in which communication and interaction between two individuals or between an individual and a computer are facilitated by digital data exchanged over computer networks. This interaction or communication can be used for a host of different purposes.

The Internet is currently the biggest network for linking computers, but cyberspace, as a concept, is independent of the Internet. Cyberspace communication began before the Internet and the World Wide Web. Cyberspace interaction and communication will continue to occur after the Internet is no longer the network of choice.

Currently, there is no generic system for identification in cyberspace. It is impossible to identify an entity or accurately tell whether an object has a specific characteristic. Digital environments have inherent differences from real space, which causes this discrepancy. When implementing an identity system for cyberspace, one needs to consider more than just the architectural nature of the system; any system chosen will have social repercussions that need to be also taken into account.

Identity is a unique piece of information associated with an entity. Identity itself is simply a collection of characteristics that are either inherent or are assigned by another. The colour of a person's hair is good or bad and whether he is attractive or not is part of a person's identity, which another person usually reviews.

Interactions done in real space inherently carry the person's identity originating the transaction. Generally, physical traits are carried along in a transaction for example, when one purchase a book from a book store, the book dealer may remember the buyer's face or build.

In this Unit, we will study the cyberspace, difference between real space and cyberspace, digital identity, cyber crimes and their prevention mechanisms.

7.1 LEARNING OUTCOMES

After studying this Unit, you should be able to:

- describe what cyberspace is;
- explain the difference between Real Space and Cyberspace;
- explain the concept of Digital Identity;
- describe Computer and Cyber Crimes;
- describe the architecture of cyberspace;
- state implications of choosing the link system; and
- list the barriers before cyberspace identity mechanism.

7.2 REAL SPACE VS CYBERSPACE

The difference between real space and cyberspace is that the essence of any digital transaction is unbundling. Ones and zeros do not inherently carry any separate information along with them; a real space transaction carries along with inseparable secondary information. Digital transmissions can only transmit; there is no secondary information encoded in the transmission unless explicitly put there. Thus, for authentication purposes, additional information needs to be carried with cyberspace transactions for identity purposes.

Providing extra information in digital communication introduces the possibility of identity theft as nothing can prevent the transmission of false identity information, or the duplication of another's identity information. The actual identity must not be transmitted along with the message; instead, a verification scheme must convince the recipient that the sender sent the message. This eliminates the need to send one's actual identity. The concept of verifying instead of revealing provides the sender with an extra layer of security.

The other point of insecurity is in the digital certificates issued to verify these characteristics. These certificates are meant to be used only by their owner. Still, if another party obtains them, that party can falsify his identity, representing him as the individual he has digital certificates.

Architecturally, we must decide how to store and use these certificates. The certificates can be stored on a smart card on a computer terminal. The certificates can be stored in an "identity server" locked via password or biometrics information and available for transmission over the Internet.

It is difficult to select, verify, or reveal portions of one's identity in real space: most forms of identification contain more information than is needed for any transaction.

The unbundling that is possible in cyberspace allows portions of identity to be disassociated and verified by a third party.

This not only creates the ability to verify via the least revealing means, but it also creates the framework for anonymous transactions. It is possible to verify the proper information without distributing the same characteristic merely. Further, cyberspace users have control over the strength of the link between their real world and the cyberidentities.

7.3 DIGITAL IDENTITY: AN OVERVIEW

A digital identity system must serve several functions:

- 1) Authentication ensuring that when a message purports to be from *Alice*, *Alice* sends it, not someone is pretending to be *Alice*.
- 2) Message integrity providing certainty that when a message arrives from *Alice*, it is the same message that *Alice* sent, not modified en route in any way.
- 3) Non Repudiation ensuring the inability of *Alice* later to deny that she sent the message and the inability of the recipient of *Alice*'s message to deny that the message was received.
- 4) Establishing digital identity architecture may have the beneficial side effect of facilitating confidentiality through encryption. The knowledge that none besides *Alice* can read a message intended for her. Before proceeding with cyber architecture, however, it is important to

examine the concept of identity itself. This section develops a working definition of identity, considers how people use their identities, and articulates why it is important to protect our identities, especially in the digital context.

7.3.1 Working Definition of Identity

It is difficult to craft a formal definition of identity. Basically, an entity's essential and unique characteristics are its identity. For example, how the system will identify this person is called Joe Jindo, where there are many Joe Jindo's around the world. These characteristics might include, among other things, the unchanging physical traits of the person, his preferences, or other people's perceptions of the individual's personality. A person's skills can also become part of one's identity.

For example, a person's identity could include the fact that he "has the ability to drive" or that he "has brown hair". Some characteristics, such as height, have one correct setting. Those traits of an individual that reflect someone else's perceptions do not have to have an absolute setting. Bob may set Alice's "is friendly" flag to true, whereas Charles may set the same flag to false. Even if Bob and Charles agree on what should be the flag's setting for Alice, Alice's own view may differ from theirs. Thus, in practice, there is a degree of fuzziness to the definition of an entity's identity, and most certainly, that how others perceive it. No two identities are the same. Each identity maps to a unique set of characteristics. Two people may share some of the same characteristics, such as being old enough to drive or having the same hair colour, but that does not mean that they have the same identity. If Joe Jindo 2 can identify himself as Joe Jindo 1, then Joe Jindo 2 can access and manipulate all the private information of Joe Jindo 1, which is called identity theft.

7.3.2 Identity as a Commodity

In today's economy, identity information often is viewed as a valuable commodity. This view of identity is worth a closer examination. Businesses desire to advertise their products to the most interesting markets and may even retool them to be more appealing to certain market segments. Knowing the preferences of individuals allows a corporation to target their products perfectly to those who would prefer and, thus, be most likely to purchase them. Making a detailed survey of an individual's preferences is very difficult, if not impossible.

Often an individual cannot specify the exact motivation for purchasing a particular product. Determining which questions to ask purchasers can be daunting from the seller's perspective. Further, despite their potential usefulness, certain questions are not likely to be answered by a purchaser. To work around this problem, businesses use identity information to proxy preferences. For example, rather than discovering the exact reason an

individual purchased a Ford Mustang, a car dealer might instead try to find out the purchaser's profession or income level.

Suppose the car dealer discovers that many of his customers who purchased Ford Mustangs are lawyers. Although the car dealer may not understand why they purchased Ford Mustangs, he can assume with some level of confidence that there is something about lawyers that leads them to purchase Mustangs instead of Cougars.

Check Your Progress: 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1. Why is identity viewed as a valuable commodity?

.....
.....

7.4 VERIFYING VS REVEALING AN IDENTITY

Cyberspace creates opportunities for identity theft. One inherent property of digital media is that it can be duplicated perfectly and easily. Exact copies of everything sent over a digital communications channel can be recorded. Consider the act of sending a signed letter to someone. In the real space, I will reveal the exact form of my signature to the recipient. Still, the difficulty of mastering the art of forgery protects me from the possibility that the recipient would begin signing letters with the sender's signature.

However, suppose we send a digital letter containing our signature's digital representation. In that case, the recipient could easily duplicate and use our signature to assume our identity when signing documents. The seriousness of this problem is highlighted when you consider that future technologies will allow extremely important identifiers, such as a retinal scan or a fingerprint, to be represented digitally. These biometrics characteristics are protected in real space because they are embedded in their physical bodies. This is lost in cyberspace. Thus, cyberspace needs a system that allows individuals to verify their identities to others without revealing to them the digital representation of their identities.

A verification system would let Bob, for example, know the identity of Alice or that she possesses a particular trait but would not give him the ability to impersonate Alice or use the trait identifier as if it was his own. In our digital letter example, Bob would be able to verify that the letter contains Alice's signature but would not let him sign the documents as Alice. Similarly, a verification that someone is of the proper age to purchase alcohol would not give the person a chance to verify this identifier anything that would allow him to represent himself as being of the proper age to purchase alcohol. Such

a system helps both parties obtain what they want out of exchanging identity information without the risk of identity theft.

7.5 CYBER AND COMPUTER CRIMES

Computer crimes can involve criminal activities that are traditional in nature, such as theft, fraud, forgery and mischief, all of which are subject everywhere to criminal sanctions. The terms computer misuse and abuse are also used frequently, but they have significantly different implications. Annoying behaviour must be distinguished from criminal behaviour in Law. As per IT Act, 2000, no description has been categorically made for computer crime and cybercrime. So till today, it is very difficult to differentiate between these two words. In relation to the issue of intent, the principle of the claim of right also informs the determinations of criminal behaviour.

For example, an employee who has received a password from an employer, without direction as to whether a particular database can be accessed, is unlikely to be considered guilty of a crime if he or she accesses those databases. So, a distinction must be made between what is unethical and what is illegal; the legal response to the problems must be proportional to the alleged activity. Common types of computer crimes are:

- Forgery;
- Fraud by system manipulation intentionally;
- Any modification to data or programs or databases; and
- Accessing computers without authorisation;

But cyber crimes are somehow different from computer crimes. Computer crime happens in physical space with or without the network. Cybercrime takes place in a virtual space through a digital environment. Cybercrimes may happen globally as there is no geographical limit for cyberspace.

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1. Give two examples of Computer Crimes.

.....

7.6 ARCHITECTURE OF CYBERSPACE

Practically cyberspace architecture for the global standard is impossible. However, certain groups of networks maintain some rules and regulations to

make minimum architecture through TCP/IP and a virtual global server system. Here some theoretical architectural choice has been described.

7.6.1 Link and No Link: An Architectural Choice

As identified earlier, any digital identification system must determine where to lie upon the continuum of anonymity and accountability; that is, a system must adopt an appropriate degree of Type II unbundling. However, within the context of law enforcement, it becomes clear that not all points along this continuum are equal. One point is very different from all the others: the point at the far end of the spectrum where there is absolutely no traceability. For the sake of clarity in our further discussion, this point will be called “nolink”. At the nolink point, the architecture exists within the digital identification, which has no mechanism for determining the link between data in cyberspace and the real world recipient or sender. The nolink point implies that there is no mandatory link between cyberspace and the real world; this does not preclude an additional, nonmandatory method of determining an identity that could be layered on the top of the nolink architecture. All other points along the spectrum will be designated as “link” points. This indicates that there is some mandatory architectural mechanism for determining the real-world identity of the sender and receiver of data in cyberspace.

Both link and nolink architecture have benefits and drawbacks associated with them. With link architecture, access to the link information can be limited, presumably, only to an appropriately regulated law enforcement agency with specific regulatory processes in place for obtaining the information. However, the immediate point is that not everyone will have access to the information contained in the architectural link; to those without access, link architecture is identical to nolink architecture. The benefit of identification is still present, but the ability to gain knowledge of the person’s real-world identity from the system’s architecture is limited to those specific bodies with access. Thus, once again, the interesting area of discussion is that pertaining to law enforcement: when can a link system effectively be used *laissez-faire*, and are there benefits able to determine a link which outweighs any corresponding drawbacks?

At all the points along the continuum, except for the extreme of one to one identity, there is a need to distinguish between “transient anonymity” and “persistent anonymity”.

With transient anonymity, no persistent link remains to the sender of the information; this is analogous to anonymous leafleting. Persistent anonymity is perhaps more useful: it allows continuity of cyber identity, generally without disclosing the real-world identity, i.e., both the sender and receiver mutually agreed and define their private communication channel in the network, which is not accessible to any other at any circumstances unless the private information of any party is not tempered or compromised. It only permits disclosure of the real world identity within a link system. In a nolink

system, continuity is preserved, but without facilitating the link. Both types of anonymity are useful in some circumstances, but persistent anonymity is likely to be more generally useful.

No Link

The benefits of a nolink system are, as mentioned above, those pertaining mostly to issues of freedom of speech and freedom of action. In the commercial domain, the wheels of capitalism are greased by the nolink architecture. People who have no fear of ever being personally associated with what they buy are far less likely to be concerned about the social norms which might have previously restricted them from purchasing a product. Unbundling facilitates the necessary degree of identification that commerce will require without necessitating the revelation of the entire real-world identity. The absence of traceability likewise assists free speech: where potential oppressors are unable to determine the sender's real-world identity, there is no danger of oppression.

Link Architecture

No link architecture provides protection from McCarthyism. But in so doing, it removes all accountability from speech. It is an architecture that completely eliminates the power of social norms, market regulation, and legal regulation to govern interaction on the Internet. Society should not overlook the more general consequences that may result from the ability to avoid accountability in all speech, especially the speech which would not be considered criminal: people may routinely, without concern, spout inaccurate and misleading information, and the responsibility may disappear even further from the moral landscape. However, the aspects that can be most clearly identified and discussed result in criminal behaviour.

Check Your Progress: 3

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1. Discuss the various types of Cyberspace Architecture.

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7.7 PREVENTING CRIMES

The issue then becomes one of the preventing crimes while simultaneously attempting to mitigate this potential “chilling effect” on free speech. The distinction between transactional and content information lies at the heart of this discussion.

Transactional information is the information regarding the sender, recipient, and other information associated with the transmission of the information, but

not regarding the content of the information. Thus, so far, the argument has centred on transactional information; however, the value of content to law enforcement must be considered: if it is absolutely necessary to have content as well as transactional information, then it will do no good to consider offering the latter without the former. If, on the other hand, transactional information without content is a tool that can be utilised, it may result in an effective compromise between the needs of law enforcement and the desires of society.

Encryption represents the single largest barrier to law enforcement obtaining content from a computer. This is a relatively unique issue to cyberspace, as handwritten and telephone encryption is relatively rare. One choice can be made with respect to encryption: allow it, without regulation, or disallow it. Disallowing encryption altogether is pragmatically different from allowing only key escrowed encryption, but for the purposes of this discussion, they are effectively the same. The overwhelming response of the government has been that encryption controls are in fact necessary, and several initiatives have been proposed to this effect; however, both the public and legal reaction to these initiatives have been negative: many organisations are resisting the degree of control which law enforcement would be given, and the Communications Decency Act was recently ruled as too general to be constitutional. In this situation, law enforcement's claims of what it needs to be effective are strongly disputed by the public: the equilibrium between the two is harder to strike in cyberspace.

7.8 IMPLICATIONS OF CHOOSING THE LINK SYSTEM

The negative implications of choosing the link system are clear: it may place an unreasonable burden on free speech. Even if it is not unconstitutional in this manner, it may simply deter people from speaking out in situations where their voices would be most useful. To convince the society that its interests in avoiding unreasonable persecution are maintained, the architectural decision to include the link must be combined with legal regulations regarding who is given sanction to disclose the link and under what circumstances such disclosure is acceptable. While the negative impacts of providing a link with all the transmitted data can never be fully accounted for, the goal of a system that provides an architectural link must be to mitigate the impact of the architecture as fully as possible.

Nolink architecture has more tangible drawbacks. Crimes can be easily planned and carried out on a system with no accountability, and there is no reason to think that they would not be. However, practical concerns such as sovereignty and providing unrestricted speech to political dissidents regardless of their government's policy on free speech may outweigh the potential societal costs. It may also be that suitable mechanisms for regulating their identity can be created in a legal or market-based way; it is hard to see how these methods would be enforceable in a costeffective manner, but the

number of criminal deviants might be small enough that the identification by law enforcement could be reasonably achieved.

A Note on Architectural Choice

It is very important whether the architecture of a digital authentication mechanism should be designed to permit traceability. Although the discussion focuses on how traceability on the Internet would meet the needs of the government in carrying out its law enforcement function, it should be noted that businesses also have an interest in the development of architecture with such a feature. Many corporations have established Intranets to facilitate communication between the various divisions of their companies. Traceability in the architecture would help the leadership of a business monitor the activities of its employees. Monitoring of this sort might be motivated by a desire to track the productivity of the individual workers or a need to ensure procedures designed to govern access to the company's sensitive information are followed. The development of the architecture for the Internet that included traceability would provide a standard that could be adopted for the internal corporate networks, without the associated research and development costs.

Aside from the caveat presented above, businessdomain interests in the use of identity do not require the architecture developers to make any fundamental architectural choices for the system. Instead, most of the concerns regarding the business arena are related to how businesses and consumers will behave in an environment using the digital authentication mechanism proposed.

Social Aspects

Community in cyberspace is based on the interaction between people. Cyberspace has an important social aspect to it that must not be overlooked. Since the ARPANET was created, its primary use has been to communicate with other people. With the advent of a faster backbone, different communication media became possible, namely, interactive communications. Community in cyberspace is based on the interaction between people.

Although a community is a group of people who interact with each other, at the basic level, it comprises a group of people who exist with each other in a common plane.

Cyberspace can be treated as a conduit touching portion of real space at key points. Ideas are passed through the conduit, and business is transacted through this conduit. Cyberspace communities are members of the global community interacting on a different plane than in real space. These members rarely interact in the real space, but they communicate through multimedia means in cyberspace, whether it be by text, image, sound, or a combination of the three. It is not possible to use the Internet without being part of this community of people; you cannot avoid being a part of the community, even if you are using the Internet as a conduit: by emailing

people, reading web pages, reading newsgroups, or doing commerce online, one has joined the cyberspace community.

7.9 ROAD TO IMPLEMENTATION

The current state of cyberspace identification mechanisms is far from identity architecture's broad, flexible potential. There is still a long way to go from the 'here' of the Internet as it exists in 1998 to the 'thereof the ubiquitous, secure identity architecture. A secure mechanism for managing and verifying digital identity is necessary for the Internet to reach its full potential. There remain ranges of hurdles to overcome before a cyberspace identity mechanism will be deployed and ubiquitous. These hurdles can best be analysed in four categories: social norms, market, legal, and architectural barriers.

Social Norms Barriers

The main social obstacle to implementing a cyberspace identification mechanism is that the general public does not recognise that there is a problem with the existing identification architecture. The general public does not understand the need for an improved, secure cyberspace identification system. Even without any effective identification mechanism, the use of the Internet for both casual and secure applications has soared, with doubledigit growth rates measured monthtomonth rather than yearto-year. While more sophisticated Internet users may recognise the need for a digital identity mechanism, these advanced users represent a shrinking percentage of the overall Internet? Community? Many people using popular Internet applications seem to be satisfied with the existing levels of security and identification. Email, for instance, is often self-identifying through the content of the message. Forged email, while easy to create in the current architecture, is not perceived to be a major problem.

Email eavesdropping, also a relatively simple technical task, has not slowed the flood of email communications. Online commerce is booming even based on systems requiring credit card numbers and the overly revealing identification that credit card numbers enable.

Market Barriers

The market barriers to implementing a secure Internet identification system stem from the difficult business economics inherent in solving this type of problem. One of the key problems is that there is a significant business model risk for companies providing identity verification solutions. In other words, it is unclear exactly how these companies can make money. In addition, economic incentives do not encourage the development of an openstandard identity infrastructure. Ultimately, the success of an open-standard identification architecture, such as our proposed system, may require government intervention in the marketplace.

Legal Barriers

The most critical legal obstacle to developing and adopting an effective digital identity mechanism is the current confusion over legal liability rules. In other words, who is responsible if someone's digital identity is misused or stolen? Who bears the cost if a digital identification mechanism is compromised? The lack of a clear legal liability regime for these two issues discourages the cyberspace identity market from emerging in the first place and from operating efficiently once it does become widespread. Legislatures may need to enact liability laws that cover digital identity before the identity infrastructure can be effectively implemented.

The appropriate liability rules must reconcile two competing principles. First, because the market for digital identity mechanisms is in its infancy, the selected liability rules must help create incentives that will drive towards the widespread adoption of secure identity infrastructure. According to this goal, the liability for identity misuse should be placed on whichever party can best induce the introduction and implementation of the identity architecture. Second, to have an efficiently operating marketplace for identity mechanisms, the selected liability rules should place liability on the party who is the "least cost avoider" of harm. Adopting this goal, liability for identity misuse should be placed on whoever is best able to avoid misuse of the digital identity. If these two goals point towards the same party, both goals can be accomplished together. However, if these two goals suggest that different parties should bear liability, then one goal or another must be made paramount, or the goals must be balanced.

Architectural Barriers

In broad terms, there are just three types of identification mechanisms. Authentication can be based on a person's shared knowledge (such as a password), a person's possession of unique information or device (such as a digital certificate), or a person's inherent unique characteristics (such as a fingerprint or other biometric).

Check Your Progress: 4

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1. What are the barriers before cyberspace identity mechanisms can be deployed?

.....

7.10 LET US SUM UP

There is no proper definition of cyberspace yet. Only some concepts have been derived. Cyberspace is the total interconnection of human beings

through networked computers and telecommunications without regard to physical geography. The difference between real space and cyberspace is that the essence of any digital transaction is unbundling. The main problems are to identify the reality. Digital Identity is the mechanism to identify the man or product through the digital environment. In the present scenario, digital identity is also often viewed as a commodity. Computer crime and cybercrime seem to be similar, but both are different. Computer crime belongs to any individual computer without an Internet connection, i.e. physically, whereas cybercrime happens in cyberspace through the Internet only.

Cyberspace architecture, which is not properly defined now, is a design in which virtual space transactions are made through the digital environment. Presently the cyberspace identification mechanism is not flexible; there are a number of barriers, for example, social norms, market, legal and architectural, before a cyberspace identity mechanism could be deployed.

7.11 FURTHER READINGS

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7.12 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. As there is no chance of physical verification of personal identity in cyberspace, cyberspace plays a crucial role in electronic identity. So, this electronic identity (called identity only) is viewed as a commodity for commercial purposes.

Check Your Progress: 2

1. Forgery, Accessing the Computer without Authorisation

Check Your Progress: 3

1. In practice, there is no specific architecture defined for cyberspace, but some theoretical concept has been yet proposed for the same, like link and nolink architecture for architecture framework.

Check Your Progress: 4

1. The cyberspace identification mechanism is not flexible; there are a number of barriers, for example, social norms, market, legal and architectural, before a cyberspace identity mechanism could be deployed

UNIT 8 CYBER LAW

Structure

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8.0 INTRODUCTION

In the era of rapid use of information communication technologies, networks, devices and services worldwide, cyberspace has emerged as a new medium of communication. According to International Telecommunication Union (ITU), International Internet bandwidth is growing rapidly worldwide, and India is among the front runner nations of the world. In India, the Internet is growing at a fast pace in cities and villages. More and more people are using mobile telephones, which has further accelerated its growth in the country, the new virtual space has thus given rise to many social and national issues and has posed new challenges to deal with technology-driven cybercrimes and protect a person's privacy and intellectual rights.

This Unit provides you with an overview of the concept of cyberspace, cybercrimes and other issues emerging from it.

8.1 LEARNING OUTCOMES

After studying this Unit, you should be able to:

- discuss the concept of cyberspace;
- explain the need for regulation of cyberspace;
- outline various issues that emerged due to cyberspace;
- describe cyber crimes and offences under Information Technology (IT) Act; and
- acquaint yourself with the law regulating cyberspace in India.

8.2 CONCEPT OF CYBERSPACE

Cyberspace represents a space created by science where various events and ideas are taking place with the help of the Internet connecting various computer systems and mobile phones.

With the advent and growth of electronic communication, the word “cyberspace” has entered into our everyday vocabulary. But what does this word signify? To a common person, cyberspace refers to a virtual area without limits to meet people and discover information on any subject through the Internet. In cyberspace, you can find the right, wrong or confusing answer to almost any question. This description of cyberspace is not 100% accurate, but it points out some of its important characteristics: non-physicality, social dimension, and functionality.

According to Rebecca Bryant, Cyberspace represents “the new medium of communication, electronic communication, which is fast outmoding, or even replacing, more traditional methods of communication. It appears, then, that space and cyberspace can count as roughly equivalent conceptual entities, at least in the sense of sharing the four common sub-concepts of place: distance, size, time and route.

Despite the differences, cyberspace is, in one way, intimately connected with the physical world. Cyberspace depends, for its very existence, on hardware and software, cables and routers it depends on physical objects existing in physical space. And, of course, this intimate connection between the two also represents a fundamental difference - physical space, if it exists, depends on nothing at all”.

8.2.1 Characteristics of Cyberspace

David B. Whittle, in his book, “Cyberspace: The Human Dimension”, has identified three characteristics of cyberspace: (1) It is not a physical location but a virtual space. (2) One needs an access device to enter into cyberspace. That means one needs some sort of physical access device (maybe a computer screen, a telephone, a terminal, etc.) with an artificial processing mechanism, such as digital computing power and/or software. And that should be joined with other access devices on a network of physical connections. (3) It enables interaction and communication between individuals and groups of individuals and their creative output, largely independent of time and space.

8.2.2 Issues Emerging From Cyberspace and The Need for Regulation

Today new communication technologies, usage of mobile phones and other communication devices are globally challenging the traditional notion of jurisdiction. It has led to the possibility of invasion of an individual's privacy. There is a need to have effective law to deal with the problems of cyber social media crimes, rumour-mongering, email spoofing, spam, cyberstalking, defamation, and various other cyber crimes as the impact of these crimes can be more than the conventional crimes. These cybercrimes are usually committed by the person using a fake identity, not readily and easily identifiable.

Further, anonymity and the ease of circulation have given rise to many social debates that demand the reconciliation of the two views, viz., freedom of speech and expression and the concern for maintaining basic civic peace and standards. Besides this, the major areas of concern that calls for strict regulation are: Management of Intellectual Property and to prevent Infringements in digital media; spread of terrorism, cross border taxation; cyber security, which is an expensive affair as the business organisations are vulnerable to data breaches leading to loss of business opportunities and therefore needed to be secured from unauthorised access, modification or removal of data/information, data theft; authentication, data protection and data privacy of the industries, individuals and government agencies; Encryption; Protection of e-consumers from the misleading advertisements of goods and services enabling them to make informed and meaningful choices. Business organisations must manage consumer information responsibly, respecting the privacy of the individuals, but this requires strict regulators restricting the exchange and use of data.

Check Your Progress: 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1) What are the characteristics of cyberspace?

.....

2) Why do we need cyberspace regulation?

.....

8.3 INTERNATIONAL AND NATIONAL CYBER LAWS

This section shall briefly discuss the international and national laws that govern cyberspace.

8.3.1 International Law

UNCITRAL Model law 1996, Model law on E-Signature, 2001 and United Nations Convention on the Use of Electronic Communications in International Contracts, 2005 constitute some international initiatives relating to cyberspace.

UNCITRAL Model law 1996 -The first Model Law on E-commerce was adopted in 1996 by United Nations Commission on International Trade and Law (UNCITRAL). The General Assembly of United Nations adopted it by passing a resolution on 30 January 1997. The prime objective of the law was to have uniformity in law relating to e-commerce at the international level and to provide equal treatment to paper-based and electronic information.

India is also a signatory to this Model law and enacted the Information Technology Act, 2000.

A model law on E-Signature, 2001(MLS) - In 2001, the Model Law on E-Signature was adopted by United Nations Commission on International Trade and Law (UNCITRAL) with the aim to enable and facilitate the use of electronic signatures by establishing criteria of technical reliability for the equivalence between electronic and hand-written signatures. The law may assist countries in establishing a modern, harmonised and fair legislative framework to address the legal treatment of electronic signatures effectively and give certainty to their status. Accordingly, India passed the Information Technology (Amendment) Act, 2008, which made necessary amendments in 2000.

United Nations Convention on the Use of Electronic Communications in International Contracts, 2005 - It was adopted on 23 November 2005 and came into force on 1 March 2013. It recognises that electronic communications play a fundamental role in promoting domestic and international trade and economic development and improving commercial activities' efficiency. It aims to provide a common solution to remove legal obstacles to the use of electronic communications in a manner acceptable to States with different legal, social and economic systems. The Electronic Communications Convention aims at facilitating the use of electronic communications in international trade by assuring that contracts concluded and other communications exchanged electronically are as valid and enforceable as their traditional paper-based equivalents.

8.3.2 National Law

Being a signatory to UNICITRAL LAW on E-commerce, the Government of India enacted 2000, which was amended in the year 2008 to implement the UNCITRAL Model Law on Electronic Signatures, 2001. Many traditional crimes which are capable of being committed with the use or aid of or through computers and technology have been brought within the definition of conventional crimes and therefore fall under the ambit of the Indian Penal Code, 1860, as amended. The Evidence Act, 1872 has been amended; section

65A and section 65B of the Indian Evidence Act, 1872, provide the Admissibility of electronic records as evidence. The Banker's Book Evidence Act, 1891 and the Reserve Bank of India Act, 1934, have also been amended in order to facilitate the collection of evidence to deal with cybercrimes or any matter connected with such crimes. The main purpose of these amendments is to address the related issues of electronic commerce, electronic crimes and evidence and to enable further regulation with regard to Electronic Fund Transfer.

For further detailed information, you may visit the website of the United Nations Commission on International Trade Law: www.unictr.org.

8.4 INFORMATION TECHNOLOGY ACT, 2000 AS AMENDED

The Information Technology Act, 2000 aims to provide legal recognition for the transactions carried out by means of electronic data interchange and other means of communications commonly referred to as "Electronic Commerce", which involve the use of alternatives to paper-based methods of communication and storage of information, to facilitate electronic filing of a document with the government agencies.

8.4.1 Electronic Signature and Electronic Records

Section 3A of the Information Technology Act provides that a subscriber may authenticate any electronic record by such electronic signature or electronic authentication technique which (a) is considered reliable and (b) may be specified in the Second Schedule.

Chapter III of the Act pertains to legal recognition of the electronic records (Section 4), an electronic signature (Section 5) and their usage in Government and its agencies (Section 6). Chapter IV lays down rules for attribution of the e-record, the mode and manner of its acknowledgement and determination of time and place of dispatch and receipt of electronic records. Section 10A provides for the validity of contracts formed through electronic means. Chapter V lays down conditions for secure electronic records and secure electronic signatures.

8.4.2 Regulation of Certifying Authorities

The provisions relating to Regulation of Certification Authorities are given in Chapter VI of the Information technology. Chapter VI deals with the appointment of a controller and other officers, functions of Controller, recognition of foreign certifying authorities, licence to issue electronic signature certificates – application, renewal, suspension of licence and procedure for grant or rejection of licence.

Functions of Controller

The Controller of Certification authority is a focal point in the Information Technology Act, who shall discharge the functions under this Act subject to

the general control and directions of the Central Government. According to Section 18 of the Information technology, the Controller may perform all or any of the following functions:

- a) Exercising supervision over the activities of the Certifying Authorities.
- b) Certifying public keys of the Certifying Authorities.
- c) Laying down the standards to be maintained by the Certifying Authorities.
- d) Specifying the qualifications and experience which employees of the Certifying Authorities should possess.
- e) Specifying the conditions subject to which the Certifying Authorities shall conduct their business.
- f) Specifying the contents of written, printed or visual materials and advertisements that may be distributed or used in respect of a Digital Signature Certificate and the public key.
- g) Specifying the form and content of a Digital Signature Certificate and the key.
- h) Specifying the form and manner in which the Certifying Authorities shall maintain accounts.
- i) Specify the terms and conditions subject to which auditors may be appointed and the remuneration to them.
- j) Facilitating the establishment of any electronic system by a Certifying Authority either solely or jointly with other Certifying Authorities and regulation of such systems.
- k) Specifying the manner in which the Certifying Authorities shall conduct their dealings with the subscribers.
- l) Resolving any conflict of interests between the Certifying Authorities and the subscribers.
- m) Laying down the duties of the Certifying Authorities.
- n) Maintaining a database containing the disclosure record of every Certifying Authority containing such particulars as may be specified by regulations, which shall be accessible to the public.

Powers of Controller

Section 24 of the procedure for grant or rejection of licence is stated. According to this Section, the Controller may, upon receipt of an application under Section 21(1), consider the documents accompanying the application and such other factors, as he deems fit, grant the licence or reject the application. However, no application shall be rejected under this section

unless the applicant has been given a reasonable opportunity of presenting his case.

Section 25 of the procedure for suspension of licence is laid down. According to this Section:

1. The Controller may if he is satisfied after making such inquiry, as he may think fit, that a Certifying Authority has-
 - a) made a statement that the application for the issue or renewal of the licence is incorrect or false in material particulars;
 - b) failed to comply with the terms and conditions subject to which the licence was granted;
 - c) failed to maintain the procedures and standards specified in Section 30;
 - d) contravened any provisions of this Act, its rules, regulations or orders; revoke the licence: Provided that no licence shall be revoked unless the Certifying Authority has been given a reasonable opportunity of showing cause against the proposed revocation.
2. The Controller may, if he has reasonable cause to believe that there is any ground for revoking a licence under the above sub-section (1), by order, suspend such licence pending the completion of any enquiry ordered by him. However, no licence shall be suspended for a period exceeding ten days unless the Certifying Authority has been given a reasonable opportunity of showing cause against the proposed suspension.
3. No Certifying Authority whose licence has been suspended shall issue any Digital Signature Certificate during such suspension.

Sections 28 and 29 of the Act provide power to the Controller or any officer authorised by him to investigate contraventions and access computers and data if he has reasonable cause to suspect any contravention of the provisions of this Act, its rules or regulations.

8.4.3 Cyber Appellate Tribunal

Section 57 of the Information Technology Act lays down provisions relating to Appeal to Cyber Appellate Tribunal.

- Any person aggrieved by an order made by the Controller or an adjudicating officer under this Act may file an appeal to a Cyber Appellate Tribunal having jurisdiction in the matter within forty-five days of receipt of the copy of the said order. However, no appeal shall lie to the Cyber Appellate Tribunal from an order made by an adjudicating officer with the consent of the parties. The Cyber Appellate Tribunal can also entertain an appeal after the expiry of the said period of forty-five days if it is satisfied that there was sufficient cause for not filing it within that period.

- The Cyber Appellate Tribunal shall give both the parties to the appeal an opportunity of being heard before passing such orders.
- The appeal filed before the Cyber Appellate Tribunal shall be disposed of as expeditiously as possible with an end eavour for final disposal within six months from the date of receipt of the appeal.

Procedure and powers of the Cyber Appellate Tribunal– Section 58 of Information Technology Act provides that the Cyber Appellate Tribunal has, to discharge its functions, the same powers as are vested in a civil court under the Code of Civil Procedure, 1908 while trying the suit. However, the Tribunal shall not be bound by the procedure laid down by Section 5 of the Code of Civil Procedure, 1908. Still, it shall be guided by the principles of natural justice and, subject to the other provisions of this Information Technology Act and its rules. The Tribunal shall have powers to regulate its own procedure, including the place at which it shall have its sittings.

8.4.4 Intermediaries

Section 2(1) (w) of the Information Technology Act defines Intermediary means any person who, on behalf of another person, receives stores or transmits that record or provides any service with respect to that record. Intermediary includes telecom service providers, network service providers; internet service providers; web-hosting service providers; search engines; online payment sites, online- auction sites, online market places and cyber cafes.

As the Internet has grown to permeate all aspects of the countries and economy, the role of Internet intermediaries in bringing together or facilitating interactions, transactions or activities between third parties on the Internet is crucial as they influence and determine access to and choice between online information, services and goods.

Duties of Intermediaries:

- According to Section 67 C of the Information Technology Act, Intermediary shall preserve and retain such information as may be specified for such duration and in such manner and format as the Central Government may prescribe. In case of intentionally contravening this duty, the Act provides that intermediary shall be punished with imprisonment for a term, which may extend to three years and also be liable to fine.
- Section 69(3) provides that the subscriber or intermediary or any person in charge of the computer resource shall, when called upon by Central Government or a State Government or any of its officers specially authorised by the Central Government or the State Government, must extend all facilities and technical assistance them. The intermediary will provide them access to (a) the computer resource generating, transmitting, receiving or storing such information; or (b) intercept,

monitor, or decrypt the information; or (c) information stored in computer resource.

- Information Technology Act also deals with blocking public access to any information through any computer resource. The intermediary has to comply with the direction issued by the Government in this regard. If the intermediary fails to comply with the direction issued, shall be punished with imprisonment for a term that may extend to seven years and be liable to a fine.

Role of intermediaries and the law

Intermediaries play an important role and act as the tools that enable users to access information and provide new opportunities for social activities and citizen participation. Their technical capacity to prevent harm by strengthening cyber security, e-consumer security and protecting privacy and intellectual property rights are very important. It has always been the concern across the world that besides certain duties and responsibilities, the intermediaries must also be given protection or exemption from legal liability that could arise due to the posting of illegal content by the users. In many countries like the USA and members of the European Union, there are attempts to provide legal protection to intermediaries from such user-generated content. Such protection is often termed as a 'safe harbour' protection. Our Information Technology Act also provides for exemption from liability of intermediary in certain cases as discussed below:

Exemption from liability

Under Information Technology Act, intermediaries are not liable for any third party information, data, or communication link made available to others in the following cases:

- a) Where the intermediary does not–
 - i) initiate the transmission,
 - ii) select the receiver of the transmission, and
 - iii) select or modify the information contained in the transmission;
- b) The intermediary observes due diligence while discharging his duties under this Act and observes such other guidelines as the Central Government may prescribe from time to time.

Liability of intermediaries

The intermediary will be liable in the following circumstances:

- a) Where the intermediary has conspired or abetted whether by threats or promise or otherwise in the commission of the unlawful Act;
- b) On being notified by the appropriate Government or its agency that any information, data or communication link residing in or connected to a

computer resource controlled by the intermediary is being used to commit the unlawful act if the intermediary fails to expeditiously remove or disable access to that material on that resource without vitiating the evidence in any manner.

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1) Define the term 'intermediaries'.

.....

2) Who can file an appeal to Cyber Appellate Tribunal?

.....

8.5 CYBERCRIMES

The term cybercrime refers to the wide range of crimes that involve computers and networks, where the computer is used as a tool to commit a crime or the computer itself is the target of a crime or incidental to a crime. The term cyber crimes, if used in a generic sense, its scope can be extended to cover many kinds of civil and criminal wrongs. Cybercrimes are committed against individuals or property; they are also committed against an organisation - Government, non Government; company; firm or group of individuals, or against the society at large.

8.5.1 Types of Computer Crimes

Computer crimes can be categorised as given below:

Conventional crimes committed through computers: There are a number of traditional or conventional crimes that are committed against individuals and their properties. Many of these crimes are now being committed with the aid of computers. Cyber defamation, cyber pornography, cyberstalking/harassment, cheating, digital forgery, theft, Internet fraud/ financial crimes including cheating, credit card frauds, money laundering, online gambling and sale of illegal articles, cyber terrorism etc. are the crimes punishable under both Indian Penal Code and Information Technology Act.

Cyber defamation (publication of defamatory statement about someone on a website or sending of emails containing defamatory information to the known contacts of the victim) is covered under section 499 of Indian Penal Code (IPC) read with section 4 of the Information Technology Act. Cyber frauds are also covered under Section 420 IPC. Digital forgery of documents is the creation of a document which one knows is not genuine and yet projects the

same as if it is genuine. Fraudulent birth certificates, ID cards, etc., are dealt with in various IPC and Information Technology Act sections. Cyberstalking, i.e. repeated acts of harassment or threatening behaviour of the criminal targeting the victim with the aid of the Internet, email, or other electronic communication devices to stalk another person, is cybercrime as well as a crime under IPC. Cyber pornography, showing of sexual acts can be dealt with under Sections 292 and 293 IPC, Sections 67, 67A and 67B Information Technology Act and Indecent Representation of Women's Act.

Crimes committed on a computer network and related to mail are technology-driven crimes like hacking/unauthorised access, email spamming, or Email spoofing. Email spamming means an illegal intrusion into a computer system and/or network. Email spamming also means sending a large number of emails to the victims as a result of which their account or mail server crashes. Email spoofing means an email that appears to originate from one source, although sent from another source. These crimes are dealt with under the Indian Penal Code and Information Technology Act.

Crimes relating to data alteration/destruction: computer vandalism, transmitting of virus/worms/Trojan horses/ logic bomb, theft of Internet hours; data diddling, salami attacks- insignificant alteration in customers account which in a single case would go completely unnoticed; phishing, etc. Data diddling is a type of cybercrime in which data is altered as it is entered into a computer system, most often by a data entry clerk or a computer virus. Phishing is a cybercrime often used to steal user data, including login credentials and credit card numbers, etc. Usually, it is committed by a person under the disguise of some trustworthy entity.

Section 43 of the Information Technology Act also provides for civil liability.

Crimes relating to violation of Intellectual property rights- The examples are the distribution of pirated software; and cybersquatting, i.e., obtaining a domain name consisting of the owner's distinctive trademark. The traditional laws for protecting intellectual property have also been applicable to the infringements taking place in digital media.

8.5.2 Cybercrimes: Some Cases

It will be useful for you to know about some cases related to cybercrimes dealing with a fake identity, defamation, cheating, and cyber pornography, publishing or transmitting obscene material in electronic form, etc.

SMC Pneumatics (India) Pvt. Ltd. v. Jogesh Kwatra, decided by ADJ, Delhi, on 13 February 2014

In this case, it was alleged that defendant Jogesh Kwatra being an employee of the plaintiff company, had sent derogatory, defamatory, obscene, vulgar, filthy and abusive emails to his employers as also to different subsidiaries of the said company all over the world with the aim to defame the company and its Managing Director. The plaintiff filed a suit for permanent injunction

restraining the defendant from doing the above said illegal acts. The Hon'ble Judge of the Delhi High Court passed an ex-parte interim injunction observing that a prima facie case had been made out by the plaintiff and consequently restrained the defendant from sending derogatory, defamatory, obscene, vulgar, humiliating and abusive emails either to the plaintiffs or to its sister subsidiaries all over the world including their Managing Directors and their Sales and Marketing departments. The defendant was further restrained from publishing, transmitting, or causing any information in the actual world as well as in cyberspace, which is derogatory or defamatory or abusive of the plaintiffs.

However, ADJ, Delhi in the judgment dated 13 February 2014, held that "this court is not in a position to accept the strongest argument on behalf of plaintiff under the circumstances in the absence of direct evidence to infer that it was the defendant in particular, who was sending these emails - the test of balance of probabilities is to be applied to the evidence available on record and not to the inferences". The issue was accordingly decided against the plaintiffs and in favour of the defendant, and the plaintiff's suit was dismissed.

State of Tamil Nadu Vs Suhas Katti, AMM Court, Egmore (CC No 4680 of 2004)

This is considered the first case in India of conviction under Section 67 of 2000. In this case, the accused, a known family friend of the victim, was interested in marrying her. However, the victim married another person, and that marriage later ended in divorce. The accused started contacting her once again, but she turned down his marriage proposal. The accused took up harassment of the victim through the Internet by posting an obscene, defamatory and annoying message about her in the yahoo message group and then forwarding emails to the victim through a false email account opened by him in the name of the victim. The posting of the message resulted in messages and phone calls from several persons to the victim, who also received phone calls from people who believed she was soliciting for sex work.

The accused was found guilty and sentenced for offences under section 67 of the Information Technology Act 2000 and 469, 509 of the IPC. The sentence included imprisonment for two years and a fine.

National Association of Software (NASSCOM), vs Ajay Sood and Others,

Decided by Delhi High Court on 23 March 2005.

The defendants were operating a placement agency involved in headhunting and recruitment in this case. In order to obtain personal data, which they could use for purposes of headhunting, the defendants composed and sent emails to third parties in the name of NASSCOM. Plaintiff filed the suit praying for a decree of permanent injunction restraining the defendants or any person acting under their authority from circulating fraudulent Emails

purportedly originating from the plaintiff of using the trademark 'NASSCOM' or any other mark confusingly similar in relation to goods or services. Prayer for damages was made in the plaint.

This landmark judgment delivered on 23 March 2005 brings the Act of "phishing" into the ambit of Indian laws even in the absence of specific legislation. The court observed that "An act which amounts to phishing, under the Indian law would be a misrepresentation made in the course of trade leading to confusion as to the source and origin of the email causing immense harm not only to the consumer but even the person whose name, identity or password is misused. It would also be an act of passing off as is affecting or tarnishing the image of the plaintiff, if an action is brought by the aggrieved party."

The defendants in the present case admitted their illegal acts, and the parties settled the matter through the recording of a compromise in the suit proceedings. According to the terms of compromise, the defendants agreed to pay a sum of Rs1.6 million to the plaintiff as damages for violation of the plaintiff's trademark rights. The court also ordered the hard disks seized from the defendants' premises to be handed over to the plaintiff, who would be the owner of the hard disks. Defendants, their servants and agents were barred from circulating fraudulent emails purportedly originating from the plaintiff or using the trade name NASSCOM or any other name/mark and address of the plaintiff amounting to tarnishing their image.

Activity-1

As discussed above, go through newspaper reports and identify some cybercrimes dealing with a fake identity, defamation, cheating, etc. Follow up the stories and analyse the outcome.

Check Your Progress: 3

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1. Explain the term 'phishing'.

.....

2. What is Data Diddling?

.....

8.6 LET US SUM UP

In this Unit, we discussed various aspects relating to Cyberlaw, such as the concept of cyberspace, the need for regulation of cyberspace; cyber crimes

and offences under the Information Technology (IT) Act; and the law regulating cyberspace in India.

Today cyberspace has emerged as a new medium of communication, a place where numbers of social and economic activities are going on, leading to new challenges and new forms of crimes. Some of these involve computers and networks, where a computer may be used as a tool to commit a crime or the computer itself is the target, or the computer may be considered as incidental to a crime. Therefore one cannot deny the role of internet intermediaries, their technical capacity to prevent harm by strengthening cyber security, e-consumer security and protecting privacy and intellectual property rights. It was discussed that the intermediaries need to observe due diligence while discharging their duties. The applicable law to govern electronic records and technology-driven crimes is Information Technology Act, 2000. It was also explained that many traditional crimes which are capable of being committed with the use or aid of or through computers and technology could also be dealt with under the Indian Penal Code, 1860; The Evidence Act, 1872, The Banker's Book Evidence Act, 1891 and the Reserve Bank of India Act, 1934, and others.

8.7 FURTHER READINGS

1. Gibson, William, *Neuromancer*, Harper Collins Publishers, London, 1995, p.67 S. SAI
2. Pavan Duggal, *Cyber Law*, Universal Law Publishing Co., Delhi, Second Edition, 2017
3. Prashant Mali, *Cyber law & cybercrimes simplified*, Cyber Infomedia, 4th edition. 2017
4. Vakul Sharma, *Information Technology Law and Practice- Cyber Laws and Laws Relating to E-Commerce*. Universal Law Publishing - An imprint of Lexis Nexis. 5th edition. 2016
5. Dr S.R. Myneni, *Information Technology Law (Cyber Laws)*, Asia Law House, 2017

8.8 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. Cyberspace is not a physical location but a virtual space. One needs an access device to enter into cyberspace. That means one needs some sort of physical access device (maybe a computer screen, a telephone, a terminal, etc.) with an artificial processing mechanism, such as digital computing power and/or software. That should be joined with other access devices on a network of physical connections. It enables interaction and communication between individuals and groups of

individuals and their creative output, largely independent of time and space.

2. Today new communication technologies, usage of mobile phones and other communication devices are globally challenging the traditional notion of jurisdiction. This has led to the possibility of the invasion of an individual's privacy. We need effective law to deal with the problems of cyber social media crimes, rumour-mongering, email spoofing, spam, cyberstalking, defamation, and various other cyber crimes as the impact of these crimes can be more than the conventional crimes. These cybercrimes are usually committed by the person using a fake identity, thus not readily and easily identifiable.

Check Your Progress: 2

1. Section 2(1) (w) of the Information Technology Act defines Intermediary as any person who, on behalf of another person, receives stores or transmits that record or provides any service with respect to that record. Intermediary includes telecom service providers, network service providers; internet service providers; web-hosting service providers; search engines; online payment sites, online-auction sites, online-market places and cyber cafes.
2. Any person aggrieved by an order made by the Controller or an adjudicating officer can file an appeal to a Cyber Appellate Tribunal having jurisdiction in the matter within forty-five days of receipt of the copy of the said order. However, no appeal shall lie to the Cyber Appellate Tribunal from an order made by an adjudicating officer with the consent of the parties.

Check Your Progress: 3

1. Phishing is a cybercrime often used to steal user data, including login credentials and credit card numbers, etc. Usually, it is committed by a person under the disguise of some trustworthy entity.
2. Data diddling is a type of cybercrime in which data is altered as it is entered into a computer system, most often by a data entry clerk or a computer virus.

UNIT 9 INFORMATION TECHNOLOGY (IT) ACT

Structure

- 9.0 Introduction
- 9.1 Learning Outcomes
- 9.2 Statement of Objects and Reasons
- 9.3 Application of the Act – The Extra-Territorial Effect
- 9.4 Electronic Signatures
- 9.5 E-Governance
 - 9.5.1 Functional-Equivalent Approach
 - 9.5.2 Legal Recognition of Electronic Records
 - 9.5.3 Legal Recognition of Digital Signatures
 - 9.5.4 Retention of Electronic Records
- 9.6 Adjudication
 - 9.6.1 Adjudicating Officer
 - 9.6.2 Cyber Regulations Appellate Tribunal
- 9.7 Penalties and Offences
 - 9.7.1 Penalties
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- 9.8 Network Service Provider Liability
- 9.9 Amendments to the Information Technology Act, 14000
 - 9.9.1 Information Technology (Amendment) Act, 14008
 - 9.9.2 Information Technology (Intermediary Guidelines and Digital Media Ethics Code) Rules, 140141
- 9.10 Amendments to Certain Statutes
- 9.11 Let Us Sum Up
- 9.12 Further Readings
- 9.13 Check Your Progress: Possible Answers

9.0 INTRODUCTION

In this Unit, we shall examine the provisions of the Information Technology Act, 14000, and its amendments in detail. In this Unit, we shall discuss the objectives for which this Act has been passed. This Unit will also discuss the extra-territorial application of the Act. This has become important because computer-related wrongs know no boundaries. A wrongful act committed in

one country may affect the computers and computer networks of not only the country where the wrong has been committed but also of other countries.

The IT Act has introduced new concepts such as “digital signature”, “e-governance”, etc. The Act gives legal recognition to electronic records and treats them at par with the paper-based system if all the safeguards are followed. We shall also discuss the adjudicatory mechanism provided in the IT Act. We shall also discuss the offences and penalties provided in the Act and how the offences under the Act be investigated. The investigation of IT-related offences is a very complicated affair. In these types of investigations, special kinds of investigation techniques are applied.

The Act also amends certain provisions of the Indian Penal Code, Indian Evidence Act etc. The objective of these amendments is to enlarge the definitions of certain offences so as to include within them the commission of these offences electronically and give legal recognition to evidence of electronic records.

9.1 LEARNING OUTCOMES

After studying this Unit, you should be able to:

- discuss the aims and objectives of the Act, i.e. what does the Act try to achieve?
- analyse the concept of electronic signature;
- discuss the provisions relating to e-governance and legal recognition of electronic records;
- discuss the process of adjudication;
- discuss the penalties and offences in case of the contravention of the Act;
- define the term and discuss network service provider; and
- describe the amendments of this Act and its criticism.

9.2 STATEMENT OF OBJECTS AND REASONS

The statement of objects and reasons of the IT Act reflects the purpose of the enactment and what it is trying to achieve. The concern of the framers of the IT Act was the need for information to be collected, stored and utilised in electronic form, which in turn would serve the dual purpose of facilitating e-commerce and inducting e-governance in the system.

Another object was clearly aimed at giving effect to the United Nations General Assembly Resolution¹, whereby the United Nations Commission adopted the Model Law on Electronic Commerce on International Trade Law. It recommended the States to give favourable consideration to the Model Law when they enact or revise their laws, *‘in view of the need for uniformity of the*

law applicable to alternatives to paper-based methods of communication and storage of information. Thus, the idea has been to make a shift from the paper-based system to the electronic system whereby the communication and storage of data would be through the electronic medium rather than on paper.

The solution is by giving a statutory mechanism to create and use digital signatures in the country. For this purpose, the required institution is created, which would be responsible for the issuance of Digital Signature Certificates and subsequent verification so that it can be used in e-commerce and e-governance. Certain ‘deeming’ provisions have been incorporated to supplement the existing laws and support them for the electronic era. The Act attempts to achieve the need for e-governance by providing for e-records. It provides statutory support to electronic records so that they can be used for the promotion of efficient delivery of government services. Cybercrimes have also been dealt with by providing punishment for certain computer-related wrongs. Finally, the Act provides for the electronic transfer of funds.

9.3 APPLICATION OF THE ACT–THE EXTRA-TERRITORIAL EFFECT

The application of the Act and its extra-territorial effect can be well understood by a conjoint reading of sections 1, 75 and 81. The Act extends to the whole of India. It also applies to any offence or contravention there under committed outside India by any person. However, an exception to this rule has been carved out in section 75 of the Act. Sub-section (1) of section 75 though in wider terms has made the Act also applies to any offence or contravention committed outside India by any person irrespective of his nationality, this sub-section has been made subject to the provisions of sub-section (14), which states that for the purposes of sub-section (1), this Act shall apply to an offence or contravention committed outside India by any person, if the Act or conduct constituting the offence or contravention involves a computer, computer system or computer network in India. In effect, if an act (amounting to an offence under the Act) has been committed and where any computer, computer system or computers which are interconnected to each other in a computer network and which is in India is also involved (which might be either as a tool for committing the crime or as a target to the crime), then the provisions of the Act would apply to such an act.

Due to the borderless connectivity of the computers through the Internet and the ease with which one can commit a cybercrime in India while physically located beyond the boundaries of the country, the Parliament has made the provisions of the Act applicable irrespective of where the accused might be physically located.

9.4 ELECTRONIC SIGNATURES

The concept of electronic signature was introduced under section 3A of the Information Technology (Amendment) Act 14008. Sec 14 (ta) of Information Technology Act 14000 had defined electronic signature as “Authentication of

any electronic record by a subscriber by means of the electronic technique specified in the second schedule and includes digital signature”.

The adoption of ‘electronic signature’ has made the Act technological neutral as it recognises both the digital signature method based on cryptography technique and electronic signature using other technologies.

Section 3A of the Information Technology Act 14000 is based on Article 6, “Compliance with a requirement for a signature” of UNCITRAL Model Law on Electronic Signatures 14001. According to the United Nations Commission on International Trade Law (UNCITRAL), electronic authentication and signature methods may be classified into the following categories:

- Those based on the knowledge of the user or the recipient, i.e., passwords, personal identification numbers (PINs), etc.
- Those are based on the user’s physical features, i.e., biometrics.
- Those are based on the possession of an object by the user, i.e., codes or other information stored on a magnetic card.
- Without falling under any of the above categories, types of authentication and signature methods might also be used to indicate the originator of electronic communication (Such as a facsimile of a handwritten signature or a name typed at the bottom of an electronic message).

Check Your Progress: 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1. Discuss the extra-territorial effect of the IT Act in your own words.

.....
.....

2. What are the electronic authentication and signatures methods categories, as per the UNCITRAL?

.....
.....

9.5 E-GOVERNANCE

Chapter III covers the area of legal recognition of certain paper-based concepts and functions in electronic form. Sections 4 to 8 provide for legal recognition of electronic records, digital signatures, use of electronic records and digital signatures in Government and its agencies, retention of electronic records, and publication of the rule, regulation, etc., in Electronic Gazette.

This chapter serves a dual purpose:

- a) It introduces the principle of functional equivalence; and,
- b) It provides the foundation to one of the averred objectives of the Act of introducing e-governance by ‘facilitating electronic filing of documents with the government agencies.

9.5.1 Functional-Equivalent Approach

Chapter III of the Act has adopted the ‘functional-equivalent’ approach. This approach is based on an analysis of the purposes and functions of the traditional paper-based requirement to determine how those purposes or functions could be fulfilled through electronic-commerce techniques. When adopting this approach in the UNCITRAL Model Law, attention was given to the existing hierarchy of form requirements, which provides a distinct level of reliability, traceability, and inalterability regarding paper-based documents. This approach singles out the basic functions of paper-based form requirements, with a view to providing criteria which, once electronic documents meet them, enable such e-documents to enjoy the same level of legal recognition as corresponding paper documents performing the same function. For example, if a contract is signed and sent as an electronic document, the chances of its reliability would be, in general situations, lesser than that of a paper-based document due to certain doubts as to its authenticity and chances of alteration of the contents. However, if the same electronic document is sent after being digitally signed by using a digital signature certificate issued by a trustworthy digital signature certificate provider, then since it would be able to perform the same functions of reliability, traceability and inalterability as a paper-based document, it would receive a legal sanction.

9.5.2 Legal Recognition of Electronic Records

Section 4 of the Act deems the fulfilment of the requirement of any information to be in writing in typewritten or printed form if such information fulfils two conditions. Firstly, such information should be rendered or made available in an electronic form (for example, in a floppy disk). Secondly, such information is accessible as to be usable for a subsequent reference. The purpose is to basically provide legal sanctity to the production of any information in electronic form. With reference to information, it means any information generated, sent, received, or stored in media, magnetic, optical, computer memory, microfilm, computer-generated micro fiche or similar device.

9.5.3 Legal Recognition of Digital Signatures

Section 5 proceeds on the functional-equivalent approach. It is based on the recognition of the functions of a signature in a paper-based environment. The following functions of a signature are considered in the UNCITRAL Guide:

(a) identifying a person; (b) providing certainty as to the personal involvement of that person in the Act of signing; (c) associating such person with the content of the document. Broadly, these being the functions of a signature, the purpose of section 5 is to merely introduce and give legal sanctity and acceptance to the use of digital signatures. It is not necessary as to what is the mode of signature; it may be paper-based or electronic. However, so long as the functions of the signature are being performed, such signatures will receive legal recognition. Section 5 of the Act states that where any law provides that any information or any other matter shall be authenticated by affixing the signature or any document shall be signed or bear the signature of any person, then, notwithstanding anything contained in such law, such requirement shall be deemed to have been satisfied, if such information or matter is authenticated by means of digital signature affixed in such manner as may be prescribed by the Central Government.

9.5.4 Retention of Electronic Records

Various statutes provide for storage of information (for example, for tax purposes or auditing/accounting, etc.). Such information is generally stored in paper-based mode. However, with the increase in computers for processing and storing information, it became imperative to provide legal sanctions to the storage of information electronically. Modern trade works through information technology and requires it to retain all the information, though generated, sent or received in electronic form, in paper-based mode would be a step back. Section 7 of the Act permits the retention of information in electronic form and gives legal recognition to electronic records retention.

9.6 ADJUDICATION

The Act provides for its own adjudicating mechanism and procedure. It appoints adjudicating officers conferring on them powers to adjudicate upon any allegations of contravention of the provisions of the Act or rules or regulations made thereunder. It also constitutes a Cyber Regulations Appellate Tribunal (CRAT) for the purpose of hearing appeals arising out of decisions of the adjudicating officer as also the Controller under various provisions of the Act.

9.6.1 Adjudicating Officer

Section 46 of the Act provides for the adjudicating officer's appointment, powers, and functions. Under sub-section (1), the Central Government shall appoint any officer not below the rank of a Director to the Government of India or an equivalent officer of a State Government to be an adjudicating officer. Such adjudicating officers should possess such experience in the field of Information Technology and legal or judicial experience as prescribed by the Central Government. The adjudicating officer is required to hold an inquiry and thereafter adjudge whether any person has committed a contravention of any of the provisions of this Act or of any rule, regulation, direction or order made thereunder. If, after providing such opportunity and on the basis of

inquiry made under sub-section (1), the adjudicating officer is satisfied that the person has committed the contravention, then he/she may impose such penalty or award such compensation as he/she thinks fit in accordance with the provisions of that section.

9.6.2 Cyber Regulations Appellate Tribunal

Chapter X of the Act contains provisions relating to Cyber Regulations Appellate Tribunal (CRAT). The Central Government, by notification, will establish one or more appellate tribunals to be known as Cyber Regulations Appellate Tribunal (CRAT). The Central Government will also, in such notification, specify the matters and places in relation to which the CRAT may exercise jurisdiction. CRAT will consist of one person only ('the Presiding Officer') to be appointed by the Central Government by notification.

In the exercise of its rule-making power under section 87 of the Act, the Central Government framed the Cyber Regulations Appellate Tribunal (Procedure) Rules, 14000 regulating the procedure to be followed in applications made to the CRAT.

Section 57 of the Act provides for an appeal to the CRAT. Sub-section (1) gives the right to appeal to any person who is aggrieved by order of the Controller or an adjudicating officer under this Act to CRAT having jurisdiction in the matter. However, this right is subject to the provisions of sub-section (14), which prohibits any appeal against any order of an adjudicating officer made with the consent of the parties. The appeal shall be dealt with by it as expeditiously as possible, and endeavour shall be made by it to dispose of the appeal finally within six months from the date of receipt of the appeal.

9.7 PENALTIES AND OFFENCES

Penalties and offences are dealt with in different Chapters in the Act. Chapter IX, which also harbours provisions relating to adjudication, enumerates the various penalties and the entailing civil consequences. Chapter XI deals exclusively with offences.

9.7.1 Penalties

Three kinds of conduct have been listed out in the Act, which would give rise to civil consequences. Firstly, any person involved in any action relating to damage to the computer, computer system, etc., under section 43 of the Act, would be liable to damages by way of compensation not exceeding one crore rupees to the person so affected. The second group pertains to failure to furnish information, returns, etc., under section 44. And finally, section 45 contains the residuary clause.

Section 43 of the Act provides a list of activities, including:

- A. Accessing or securing access to a computer, computer system or computer network. This, in effect, refers to unauthorised access.

- B. Downloading, copying, or extracting any data, computer database, or information from such a computer, computer system, or computer network, including information held or stored in any removable storage medium. This means data theft and would also include acts of copyright infringement like downloading music.
- C. Introducing or causing to be introduced any computer contaminant or computer virus into any computer, computer system or computer network.
- D. Damaging or causing damage to any computer, computer system or computer network, data, computer database or any other programmes residing in such computer, computer system or computer network.
- E. Disrupting or causing disruption of any computer, computer system or computer network.
- F. Denying or causing the denial of access to any person authorised to access any computer, computer system or computer network by any means.
- G. Providing any assistance to any person to facilitate access to a computer, computer system or computer network in contravention of the provisions of this Act, rules or regulations made thereunder. This is a facet of hacking.
- H. Charging the services availed of by a person to another person's account by tampering with or manipulating any computer, computer system or computer network. This refers to the theft of Internet hours.

Confiscation of computer, computer system, floppies, compact disks, tape drives or any other accessories in respect of which of any provision of this Act, rules, orders or regulations has been or is being contravened, can be resorted to under section 76.

9.7.2 Offences

Chapter XI of the Act enumerates the various acts which constitute an offence under the Act along with the punishment, be it either imprisonment or fine or both.

In case of offences committed by companies, such persons who, at the time the contravention was committed, was in charge of, and was responsible to, the company for the conduct of the business of the company as well as the company, will be, under sub-section (1) of section 85 of the Act, guilty of the contravention and shall be liable to be proceeded against and punished accordingly. However, suppose such a person proves that the contravention took place without his/her knowledge or that he/she exercised all due diligence to prevent such contravention. In that case, he/ she shall not be liable to punishment. Sub-section (14) of section 85 also deems a director, manager, secretary or any other officer of the company to be guilty of

contravention and liable for punishment if it is proved that the contravention has taken place with the consent or connivance of, or is attributable to any neglect on the part of such person. 'Company', for the purpose of this section, has been explained to mean anybody corporate and includes a firm or other association of individuals. 'Director', in relation to a firm, would mean a partner in the firm.

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1. What is the equivalent functional approach? Discuss how it is adopted in the Act with respect to digital signature and electronic records.

.....

2. Give a brief account of the powers and functions of the adjudicating officer and the CRAT in your own words.

.....

3. Discuss the provisions of the IT Act 14000 relating to penalty.

.....

9.8 NETWORK SERVICE PROVIDER LIABILITY

The issue of Network Service Providers has gained importance with the increase of offences being committed via the Internet, especially in the area of copyright infringement. They are being held up for abetting the offence by providing infrastructural facilities that help the offender commit the offence. However, section 79 of the Act provides for certain cases where they will not be liable to provide immunity to them. In case of any allegation of liability under the Act, rules or regulations against a Network Service Provider for any third party information or data made available by him/her, he/she shall not be liable if he/she proves that the offence or contravention was committed without his/her knowledge or that he/she had exercised all due diligence to prevent the commission of such offence or contravention. 'Network service provider', for the purpose of this section, has been explained to mean an intermediary. 'Third party information' is given to mean any information dealt with by a network service provider in his/her capacity as an intermediary.

To take an example, if A is hacking B's computer and using the network services provided by Z, a network service provider, then, to the extent that Z is

able to prove that the offence was committed without his/her knowledge or that he/she had exercised all due diligence to prevent the commission of such offence, he/she will be saved from any liability by virtue of section 79 of the Act.

9.9 AMENDMENTS TO THE INFORMATION TECHNOLOGY ACT, 14000

9.9.1 Information Technology (Amendment) Act, 14008

A major amendment of the IT Act 14000 was made in 14008. The amendment was passed on 1414 December 14008 without any debate in Lok Sabha and by the Rajya Sabha on the next day. President Pratibha Patil signed the proposed amendment into law on 5 February 14009. The IT (Amendment) Act 14008 aimed at tightening procedures and safeguards for monitoring and interception of data to prevent cybercrimes. Besides monitoring and interception, the amended Act also deals with the Indian Computer Emergency Response Team (ICERT) appointment, which deals with computer security and situations arising from cyber-attacks. Additionally, provisions have been made addressing pornography, child porn, cyber terrorism and voyeurism. The following amendments (Sections and Provisions) were made:

- a) **Section 66A:** Section 66A of the IT (Amendment) Act, 14008 prohibits sending offensive messages through a communication device (i.e. through an online medium). The types of information these covers are offensive messages of a menacing character, or a message that the sender knows to be false but is sent for the purpose of “causing annoyance, inconvenience, danger, obstruction, insult, injury, criminal intimidation, enmity, hatred, or ill will”. One can face imprisonment up to 3 years along with a fine under this section.
- b) **Section 67 and 67A:** Section 67 and 67(A) of the IT (Amendment) Act, 14008 prohibits sending of obscene and sexually explicit content through a communication device, respectively.
- c) **Section 69:** Section 69 has been redrafted, enabling Government agencies to intercept, monitor or decrypt any electronic information with the help of subscribers, intermediary or person in charge of computer resources.
- d) **Section 69A:** Section 69A of the IT (Amendment) Act, 14008, allows the Central Government to block content where it believes that this content threatens the security of the State; the sovereignty, integrity or defence of India; friendly relations with foreign States; public order; or to prevent incitement for the commission of a cognisable offence relating to any of the above. The Government has to adhere to a set of procedures and safeguards when doing so have been laid down in what has become known as the Blocking Rules.

- e) **Section 79:** Section 79 of the Information Technology (Amendment) Act, 14008 regulates the liability of a wide range of intermediaries in India. The section came in the limelight mostly because of the Intermediary Guidelines Rules, or IT Rules, which were made under it. The IT Rules constitute a move towards the privatisation of censorship in India.

Criticism of the Information Technology (Amendment) Act, 14008

IT (Amendment) Act, 14008 faced criticism in the form of restriction of free speech, constitutionality of the amendment, surveillance and data privacy. In lieu of Section 66(A), which was challenged multiple times, on 14 March 2015, the Supreme Court of India verdict that Section 66A is entirely unconstitutional. The Court said that Section 66A of IT Act 14000 “arbitrarily, excessively and disproportionately invades the right of free speech” provided under Article 19(1) of the Constitution of India. Similarly, the ban on Chinese apps based on Section 69A has been criticised for possibly being in conflict with Article 19(1)(a) of the Constitution of India, ensuring freedom of speech and expression to all, as well as possibly in conflict with WTO agreements. But the Court turned down a plea to strike down sections 69A and 79 of the Act, which deals with the procedure and safeguards for blocking certain websites.

9.9.2 Information Technology (Intermediary Guidelines and Digital Media Ethics Code) Rules, 140141

The Information Technology (Intermediary Guidelines and Digital Media Ethics Code) Rules, 140141 is secondary or subordinate legislation that suppresses India’s Intermediary Guidelines Rules 14011. The IT Rules, 140141, have stemmed from Section 87 of the IT Act, 14000 and are a combination of the draft Intermediaries Rules, 14018 and the OTT Regulation and Code of Ethics for Digital Media.

The IT Rules, 140141 aim to serve a dual purpose: (1) increasing the accountability of the social media platforms (such as Facebook, Instagram, Twitter etc.); and (14) empowering the users of social media by establishing a three-tier redressal mechanism for efficient grievance resolution. The IT Rules 140141 are to be governed as:

1. Guidelines Related to Social Media to Be Administered by Ministry of Electronics and IT
2. Digital Media Ethics Code Relating to Digital Media and OTT Platforms to Be Administered by Ministry of Information and Broadcasting

The salient features of the rules are:

- The IT Rules 140141 aim to empower ordinary users of social media platforms and OTT platforms with a mechanism for redressal and

timely resolution of their grievances with the help of a Grievance Redressal Officer (GRO) who should be a resident in India.

- Special emphasis has been given to the protection of women and children from sexual offences, fake news and another misuse of social media.
- The rules stress the point that online content publishers and social media intermediaries should follow the Constitution of the country and subject themselves to domestic laws.
- Identification of the “first originator of the information” would be required in case of an offence related to the sovereignty and integrity of India. A Chief Compliance Officer, a resident of India, also needs to be appointed, and that person shall be responsible for ensuring compliance with the Act and Rules. A monthly compliance report mentioning the details of complaints received and action taken on the complaints would be necessary.
- The OTT platforms, online news and digital media entities, on the other hand, would need to follow a Code of Ethics. Under the new rules, OTT platforms would be called ‘publishers of online curated content’. They would have to self-classify the content into five categories based on age and use parental locks for age above 13 or higher. They also need to include age verification mechanisms for content classified as ‘Adult’.

Criticism of the Information Technology (Intermediary Guidelines and Digital Media Ethics Code) Rules, 140141

IT Rules, 140141 have faced criticism on various counts:

- Social media companies like WhatsApp have expressed apprehensions about the provisions in the new rules, which require them to identify traceability when required to do so by authorities. They contend that this could possibly lead to the breaking of end-to-end encryption, which in turn can compromise users’ privacy.
- Rule 4(14), which makes it mandatory for every social media intermediary to enable tracing of originators of information on its platform, purportedly in furtherance of Section 69 of the IT Act, violates Article 19(1)(a) (Freedom of Speech and Expression).
- The creation of a grievance redressal mechanism through a governmental oversight body (an inter-departmental committee constituted under Rule 14) amounted to excessive regulation.

9.10 AMENDMENTS TO CERTAIN STATUTES

The Act, to further the acceptance and use of documents, evidence, and transfer of funds through electronic means, has amended the Indian Penal

Code, Indian Evidence Act, Bankers' Books Evidence Act and Reserve Bank of India Act vide the First, Second, Third and Fourth Schedule respectively. As the Act proposes such heavy induction of use of electronic means for documents and signatures, as also governance, it became necessary to also amend certain penal statutes to bring it on par with the offences relating to or committed with the help of such electronic means. Many of such offences have already been enumerated in the Act itself. However, such offences relate to a new category that has emerged with the use of computer technology like hacking, damage to computer systems, etc. There is another set of offences that were already on the statute books but with the use of electronic means have taken a new dimension, and their scope needs to be further widened by appropriate amendments in such statutes. This is what the amendments made by the Act purport to achieve.

Check Your Progress: 3

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1. Discuss the IT (Amendment) Act, 14008, in your own words. Explain the criticism faced by the amendment from a legal perspective.

.....

2. Discuss the IT Rule, 140141, in your own words. Explain the criticism faced by the amendment from a legal perspective.

.....

9.11 LET US SUM UP

In this Unit, we have examined in detail the objects and reasons for the IT Act, the applicability of the Act, i.e. the extra-territorial application of the Act, provisions relating to digital signatures, e-commerce and e-governance. This part of the IT Act deals with the recognition of the electronic record and its legalisation as an alternative to paper-based records.

The Act aims to give legal recognition to the information collected, stored, and utilised electronically to facilitate electronic commerce and e-governance. The Act also gives legal recognition to electronic signatures and provides for its issuance. It also provides a controlling mechanism to check the abuse of digital signatures. The Act adopts the equivalent functional approach, i.e. if the electronic records satisfy the same level of reliability as the paper document, it should be given the same recognition as the paper-based record.

We have also discussed the adjudicatory mechanisms provided in the IT Act, 14000. Furthermore, this Unit also emphasises the offences and penalties provided for in the Act, including the liability of the service providers. Finally, we have also examined the amendments made in the IT Act 14000 in the form of the IT (Amendment) Act, 14008 and IT Rules, 140141. We have also briefly looked at the amendments made by the IT Act, 14000 and the purpose of these amendments.

9.12 FURTHER READINGS

1. Information Technology Act, 14000
2. Information Technology (Amendment) Act, 14018
3. Information Technology (Intermediary Guidelines and Digital Media Ethics Code) Rules, 140141
4. Manohar Parrikar Institute for Defence Studies and Analyses. <https://www.idsa.in/idsacomments/it-rules-140141-dbhattacharya-0406141>
5. Press Information Bureau. <https://pib.gov.in/PressReleaseDetailm.aspx?PRID=1700749>
6. PRS Legislative Research. <https://prsindia.org/billtrack/the-information-technology-intermediary-guidelines-and-digital-media-ethics-code-rules-140141>

9.13 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. The application of the Act and its extra-territorial effect can be well understood by a conjoint reading of sections 1, 75 and 81. The Act extends to the whole of India. It also applies to any offence or contravention there under committed outside India by any person. Due to the borderless connectivity of the computers through the Internet and the ease with which one can commit a cybercrime in India while physically located beyond the boundaries of the country, the Parliament has made the provisions of the Act applicable irrespective of where the accused might be physically located.
2. According to the United Nations Commission on International Trade Law (UNCITRAL), electronic authentication and signature methods may be classified into the following categories:
 - Those based on the knowledge of the user or the recipient, i.e., passwords, personal identification numbers (PINs), etc.
 - Those are based on the user's physical features, i.e., biometrics.

- Those are based on the possession of an object by the user, i.e., codes or other information stored on a magnetic card.
- Without falling under any of the above categories, types of authentication and signature methods might also be used to indicate the originator of electronic communication (Such as a facsimile of a handwritten signature or a name typed at the bottom of an electronic message).

Check Your Progress: 2

1. Chapter III of the Act has adopted the 'functional-equivalent' approach. This approach is based on an analysis of the purposes and functions of the traditional paper-based requirement with a view to determining how those purposes or functions could be fulfilled through electronic-commerce techniques. This approach singles out the basic functions of paper-based form requirements, with a view to providing criteria which, once electronic documents meet them, enable such e-documents to enjoy the same level of legal recognition as corresponding paper documents performing the same function.
2. The Act provides for its own adjudicating mechanism and procedure. It appoints adjudicating officers conferring on them powers to adjudicate upon any allegations of contravention of the provisions of the Act or rules or regulations made thereunder. It also constitutes a Cyber Regulations Appellate Tribunal (CRAT) for the purpose of hearing appeals arising out of decisions of the adjudicating officer as also the Controller under various provisions of the Act.
3. Three kinds of conduct have been listed out in the Act, which would give rise to civil consequences. Firstly, any person involved in any action relating to damage to the computer, computer system, etc., under section 43 of the Act, would be liable to damages. The second group pertains to failure to furnish information, returns, etc., under section 44. And finally, section 45 contains the residuary clause. Section 43 of the Act provides a list of activities which, if carried out by any person without the permission of the owner or any other person who is in charge of a computer, computer system or computer network, would cause such person who is carrying out the Act to be liable to pay damages by way of compensation not exceeding one crore rupees to the person so affected.

Check Your Progress: 3

1. The IT (Amendment) Act 14008 aimed at tightening procedures and safeguards for monitoring and interception of data to prevent cybercrimes. Besides monitoring and interception, the amended Act also deals with the Indian Computer Emergency Response Team (ICERT) appointment, which deals with computer security and situations arising from cyber-attacks. Additionally, provisions have been made addressing pornography, child porn, cyber terrorism and voyeurism.

Criticism: *Restriction of free speech, the constitutionality of the amendment, surveillance and data privacy.*

2. The IT Rules, 140141 aim to serve a dual purpose: (1) increasing the accountability of the social media platforms (such as Facebook, Instagram, Twitter etc.); and (14) empowering the users of social media by establishing a three-tier redressal mechanism for efficient grievance resolution. Rules to be administered by the Ministry of Electronics and IT include the due diligence required of intermediaries and the grievance redressal mechanism. Rules to be administered by the Ministry of Information and Broadcasting include a code of ethics, a self-classification system and an oversight mechanism.

Criticism: *Violation of Freedom of Speech and Expression, Right to Privacy and Excessive Regulation*



UNIT 10 DATA JOURNALISM

Structure

- 10.0 Introduction
- 10.1 Learning Outcomes
- 10.2 Historical Background
 - 10.2.1 The Indian Experience
 - 10.2.2 Data Surveillance: The Dark Side of the Internet
 - 10.2.3 Wiki Leaks and the Gorilla War on Data Secrecy
- 10.3 Importance of Data Journalism
- 10.4 Basic Course Requirements of Data Journalism
- 10.5 Aggregators and Algorithms
 - 10.5.1 Data Regulation and Protection as a Right of Citizens
- 10.6 HDRS and Data about Human Development
 - 10.6.1 Annual Status of Education Reports (Aser)
 - 10.6.2 India Exclusion Reports
- 10.7 Times-Series Data
 - 10.7.1 Following Exercises May Prepare Students to Look At the Time-Series Data Critically
- 10.8 Simplifying the Challenges of Data Journalism
 - 10.8.1 Some Important Indian Websites/ Outlets Useful For Data Journalism
 - 10.8.2 Important International Web Resources on Data Journalism
- 10.9 Let Us Sum Up
- 10.10 Key Words
- 10.11 Check Your Progress: Possible Answers

10.0 INTRODUCTION

With the transformation of the brick and mortar economy into a digital economy, there has been a change in the way journalism is done. Data journalism is among the most significant changes visible and inevitable today. Data is described, arguably, as the ‘new oil’ with the connotation that whoever controls this precious resource will control the world in the coming years. Although the use of data, surveys and computer-assisted reporting (CAR) for journalism has been around for decades, journalists have started using new technologies to access seriously large amounts of data relatively recently. Today, the new and emerging technologies are helping both writers and readers to “understand data, explore it, and act upon the insights derived from it”¹ to make sense of the world much better than ever before. Data journalism, as we understand it today, takes journalists and their audiences from the mere numeracy to visualisation of the

larger picture and beyond, through a large-scale knowledge mobilisation, and an understanding of socio-economic trends and patterns.

10.1 LEARNING OUTCOMES

At the end of the course, students should be able to:

- explain the meaning and importance of data and its connotations in the digital world such as public domain, privacy and data protection;
- familiarise themselves with the power of data in a country's development and also in various democratic, colonial and neoliberal contexts. Use time-series data about budgets, census, economic growth, human development, education and health;
- describe concepts like Algorithms, Artificial Intelligence (AI), computerised aggregation and data surveillance;
- combine the ability to use data with narratives and storytelling techniques and think critically about the strengths and weaknesses of data use;
- analyse and visualise data (including spreadsheets and computer other assisted programmes) by using software packages and applications;
- assess how governments, industry and different institutions may be collecting and deploying time-series data and their use in public life; and
- describe the debate around open source data and unauthorised leaks with or without the public interest.

10.2 HISTORICAL BACKGROUND

The newsrooms of the eighties and the nineties assigned statistical analysis to business or research departments but data analysis as a tool for basic journalism was missing. Prior to data journalism, computer-assisted reporting (CAR) existed since the 1950s. CAR used relatively modest amounts of data for analysis and understanding patterns with the help of database software.² In 1952, American news network CBS used CAR for the first time to predict the result of the US Presidential elections. The US investigative journalists since the 1960s started analysing databases of public records with scientific methods so as to create their own stories.³ Investigations carried out by Phil Meyer of the Washington Bureau of the Free Press and his team used data to find that the Detroit riots were prompted by lack of jobs, poor housing, crowded living conditions, and police brutality.⁴ The use of social science survey methods coupled with CAR gave rise to precision journalism.⁵

10.2.1 The Indian Experience

In India the use of big data in journalism was historically through departments like the National Sample Survey Office (NSSO) under the Ministry of Statistics and Programme.

Implementation and the Census of India which has collected, collated and analysed big data every ten years since 1872. After Independence in 1947, the task which was started by the colonial rulers was continued by the Registrar General and the Census Commissioner of India under the Ministry of Home Affairs. Successive Indian Governments have also followed the tradition of releasing extensive and reliable data about the state of the economy in the form of the Economic Survey just before the union budget. The Economic Survey and the presentation of the Union Budget in Parliament have always been sources of intense debates in the Indian media (Including the Railway budget until the 92-year old practice was discontinued after the two budgets were merged in September 2016). The Reserve Bank of India also releases data through its periodic and annual reports which are thoroughly analysed in the economic papers.

So, what has changed since then? Well, technology has made it a lot easier to use machine-readable data for more accurate and longer-term analysis. Also, earlier data analysis was assigned strictly to a handful of specialists and everyday journalism rarely deployed large amounts of data for routine work, with the notable exception of elections. Among the first users of big election data in India were David Butler, Ashok Lahiri and Prannoy Roy (1989) who used the Election Commission data of 37 years (from 1952 to 1989) for a seminal book and for extensive analysis in the media.⁶

However, in today's newsrooms, a certain amount of familiarity with data analysis is required for all departments, including the desk, reporters, senior editors and even designers and illustrators. It is also common for modern newsrooms to conduct or outsource opinion surveys or demographic analysis with the use of specialised fields like psephology and urban geography with enormous amounts of data. Data journalism is fast emerging as a tool of investigating journalism, and a means for demanding transparency and accountability in governance. Modern mass media cannot fulfil its watchdog function without harnessing and analysing data. The whole issue of data for analyses of trends is also opening up new debates about privacy and data protection. A data journalist can tell a complex story quite simply and convincingly by using engaging and colourful infographics. The idea is to show with the help of data what the naked eye cannot see or what cannot be described or understood through quotes from usual sources of news stories.⁷

10.2.2 Data Surveillance: The Dark Side of the Internet

Internet is often called a democratic space or a real level playing field because it does not differentiate between the rich and the poor users. True that the digital divide puts poorest of poor people in the global South at a huge disadvantage but that does not take away the fact that the Internet does not differentiate (or discriminate) between individual users. However, it is increasingly being argued that the democratic nature of the Internet could well be, at best a myth and at worst a ruse, for promoting predatory market forces.

Yasha Levin (2018) argues in "Surveillance Valley: The Secret Military History of the Internet" that the Internet was in fact built by the powerful (US)

government to spy on the citizens. The author believes that the manipulation of personal data of the unsuspecting users is quite central to the political economy of the powerful Western countries like the US. Looking at data from the perspective of the new and emerging tech-industry giants like the Facebook, Google and Twitter and marketing companies like the Amazon, he argues that the business of these companies is to spy on their users by collecting a variety of data in order to maximise their own profits.⁸ Bringing in the emergence of a new military-digital-complex the author asserts that the military and the industry often complement one another. Nick Couldry and Ulises A Mejias (2018) argue that the ‘capture and processing’ of data (social or personal) unfolds through new “data relations” which may be understood through new forms of data colonialism.⁹ That is why most governments in all parts of the world are getting incrementally more circumspect about protecting the data of their individual citizens from predatory digital invaders.

10.2.3 Wiki Leaks and The Gorilla war on Data Secrecy

The control over data has witnessed some pitched battles between people who want to store classified data, mainly governments and big multinational corporations, and those who want to declassify them in public interest. The data in high demand can be about illegal bank accounts maintained abroad by the rich and powerful or data about crucial public information intentionally kept away from the citizens. The subject has serious repercussions for the future of journalism, data secrecy and the privacy of citizens. It also has strong ethical dimension because the unauthorized sharing of classified data may often be a crime in many countries. The opponents of wholesale data leaks believe it to be a form of theft, its supporters say just like sting journalism, this form of data journalism may be justified against the public interest it serves. What works in favour of the citizens are a variety of anti-secrecy laws such as the Right to Information Act, Whistle-blower Protection Act or the Data Protection Act.

At the forefront of the war on secrecy is the WikiLeaks which is a global non-profit organisation dedicated to leaking classified information which it believes should be in public domain. Founded in 2006 by Julian Assange, the organisation promotes data journalism by unearthing and analysing large databases regarding corruption, unethical practices, illegal secret operations and covert wars etc. One of the biggest data leaks after Assange was done in 2013 by another computer scientist and a former employee of the Central Intelligence Agency (CIA) Edward Snowden. The hacker, who is often described as a traitor (because of unauthorised leaks), a dissident or a whistle-blower (because of the nature of information leaked, ostensibly in public interest), has disclosed numerous global surveillance programmes of the US Government and its covert, often questionable, foreign-policy operations.

According to the Wikileaks’ website, it has published over 10 million documents and related analysis so far. WikiLeaks is intensely disliked by many powerful governments but it is also a recipient of some of the world’s most coveted journalism awards. Der Spiegel, one of the most influential German publications, quotes Julian Assange where he describes Wikileaks as “... a giant

library of the world's most persecuted documents. We give asylum to these documents, we analyse them, we promote them and we obtain more.”¹⁰ The organisation claims to have “contractual relationships and secure communications paths to more than 100 major media organisations from around the world. This gives WikiLeaks sources negotiating power, impact and technical protections that would otherwise be difficult or impossible to achieve.”¹¹

10.3 IMPORTANCE OF DATA JOURNALISM

According to a report from Tow Center for Digital Journalism, data journalism is the “application of data science to journalism, where data science is defined as the study of the extraction of knowledge from data”.¹² In short, data journalism is about narrating stories or analysing social and political processes around us with the use of numbers or finding stories/ patterns in numbers. The use of humongous amounts of data has been made possible by computerisation, new and emerging technologies, digital storage devices, and the opportunities of easy sharing through the Internet and multiple public domains.

Due to the explosion of data which is generated by various sectors including government, wider society, industry, academia, individual citizens, social media and the research community, both the context and scope of data journalism have expanded in the recent years. One can observe that data journalists often use free, powerful online tools and open source software to collect, clean, and publish data in interactive features, mobile apps, and digital maps.¹³ The availability of free and open source data in a convenient and modifiable form too has lent a helping hand to data journalists. Unlike in the past, when the global trends took a long time to catch up with the developing countries, data journalism is already an accepted phenomenon all over the world, including in India.

10.4 BASIC COURSE REQUIREMENTS OF DATA JOURNALISM

Data journalism needs systematic understanding of certain specific areas in the form of building blocks. If the students are comfortable with the idea of numeracy and develop basic skills of making sense of spreadsheet-based programmes, they will be able to use numbers for deeper analysis of things. The course will prepare students to learn the basic proficiency in cleaning data for their own understanding, analysing it for interpretation in the form of conclusions. In the end, they should be able to interpret quantitative academic research, tables rows and columns or from specialised programmes like Excel, spreadsheet programmes, Google Sheets and advanced visualization techniques.

The following steps can be seen as the building blocks of data journalism:

1. Gathering of data and distinguishing between reliable and unreliable sources of data. Developing an understanding of the public domain and techniques to access publicly available data.
2. Use of spreadsheets and various popular software programmes for digital number crunching. This is more of a skill which can be learnt in

simple steps and with classroom-based practical experiments. As time passes, newer and better tools of data presentation and analysis will be available, including methods using artificial intelligence (AI) with the help of mobile phone Apps.

3. Visualising data as part of a story. Too many numbers in a story can burden the narrative and make it stodgy and unreadable. However, imaginative use of charts, tables and graphics can increase the depth and gravitas of the story. This would require a team work but the reporter must be able to first visualise a story in which data, narratives and even digital maps and pictures can be combined.
4. The Editors and senior journalists must be able to question 'data-ism' where data takes over human element in a news story. As a tool of empowerment, data should be able to enhance journalistic insights and intuitions rather than creating confusion. It is equally important for the user of data to understand how incorrect, insufficient data can create false paradigms and how to differentiate between good, bad, biased and motivated data.

Check Your Progress: 1

Note: 1) Use the space given below for your answer

2) Compare your answer with those given at the end of this unit.

1. How did Data journalism begin and what are the important changes it has undergone in the digital world?

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10.5 AGGREGATORS AND ALGORITHMS

In the light of the busy lifestyles, ever increasing dependence on the Internet, and focus on getting personalised information, Content Aggregators play a significant role as purveyors of news and information on the go. News Aggregators offer an automated platform which deploy search engine optimization tools to collate and disseminate the news that has been reported by other sources. The stories which are "trending" on top of aggregated platforms are considered to be good and many news organisations compete with one another to get the top slot on, say, Google News.

In terms of computing, news aggregators (also called feed aggregators) amass pieces of information in the form of text, audio-video clips and pictures etc and present them in friendly formats to existing and potential users in order to save their time and effort. To make the news more and more relevant to the readers, the aggregators often collect personal Data about the potential and real users, their lifestyle, web usage and search histories, social media linkages, personal

interests and hobbies, age and income groups, neighbourhoods and travel profiles. Specialised algorithms are then deployed to mimic personal search for every user. Such enormous amounts of data, its collection and use, have grave implications on the future of journalism and social ethics. The data thus collected can not only be used for profiling users and selling them to advertisers but it can also be deployed for political campaigning, targeting propaganda and for manipulating the media and the elections.

10.5.1 Data Regulation and Protection as a Right of Citizens

We know from the above discussion that it is common for tech companies to do regular and real-time data snooping and the use of algorithms to customise information for individual users. It is precisely for this reason that data protection laws are required. Data protection in India is poor mainly because it is relatively new area for citizens as well as governments. However, it is important for data journalists to understand the need and ramifications of data protection. It has also become important in the light of widespread complaints of cybercrimes and data (and identity) theft. Personal data of the citizens is often stolen from the banks and insurance companies. It is often seen that even government controlled data is sometimes leaked or stolen. One of the biggest problems is the absence of an effective framework of data protection.

In Aug. 2017 the Supreme Court of India made privacy a fundamental right through a landmark order. The Apex Court warned against the threat to privacy from both state and non-state actors, asking the government to put in place “a robust regime for data protection.” The existing data protection laws (i.e. the Information Technology Act 2000 and its rules framed in 2011) cover only corporates, not government entities, and only sensitive personal data, such as medical history, biometric information etc.¹⁴ Ultimately, there is lack of recourse for citizens – breach of personal information is not ground enough for seeking redress, except in case of financial loss.¹⁵ The absence of a regulator continues to be a key weakness of the system. Justice BN Srikrishna committee set up in 2017 to suggest an effective data protection regime proposed Personal Data Protection Bill 2018. It is hoped that the data protection will be an important part of citizens’ rights in the future.

10.6 HDRS AND DATA ABOUT HUMAN DEVELOPMENT

Human Development Reports (HDRs) are published by UNDP annually to evaluates the countries on the basis of their performance on the Human Development Indicators (HDI): Life Expectancy, Knowledge and Income (per capita income) of the population. A composite index is formed from ranking the countries under the three indicators and a combined list of final ranking of the countries is published with HDR for which data is collected round the year. As per the Global Human Development Report 2016: Human Development for Everyone, India is ranked at 130 among the 188 countries. HDR comes up with open-ended, sustainable solutions to increase human development through contemporary means. The HDRs

were later published in the form of National HDRs, State HDRs and even district HDRs (for selected districts) which collate enormous amount of data about human indicators.¹⁶ On the basis of these, a country can conduct self-assessment of their performance on HDI and frame constructive policies to enforce improvement. State-level HDRs have also become a fundamental part of planning and political discourse for the majority of the Indian states in last 20 years.

10.6.1 Annual Status of Education Reports (ASER)

Annual Status of Education Report (ASER) is an autonomous research and assessment unit, established by Pratham, a civil society organisation that engages in estimating children's schooling and learning status at district, state and national levels. ASER survey reaches almost every rural district in the country and covers more than 15,000 households and 650,000 children each year¹⁷. ASER is an annual survey that aims to provide reliable estimates of children's enrolment and basic learning levels for each district and state in India. ASER has been conducted every year since 2005 in all rural districts of India. It is the largest citizen-led survey in India¹⁸. ASER is trying to generate actionable evidence and on the basis of that, solutions to deal with the situation can be identified. This evidence is derived on a variety of data collated from surveys, assessments, evaluations and other research activities in education and other social sectors. For instance, the 2017 survey examines what the students do, what they know, and what they want besides the school enrolment & learning abilities.

10.6.2 India Exclusion Reports

India Exclusion Report (IXR) is a multi-disciplinary report published by Centre for Equity Studies. It stands as a platform for social scientists, policy researchers, scholars, etc. to promote inclusion in accessing rights and government services. The IXR report often discovers inconsistencies between the needs of people and the allocation of resources by the government. The IXR, 2016 is the third edition of the report that highlights the systemic exclusion of certain groups, despite several declarations by the government. The report also looks at the impact of economic policies from the perspective of the vulnerable sections such as the Scheduled Castes, Scheduled Tribes, the minorities, women and persons with Disabilities, who were historically underprivileged. The report includes other vulnerable groups such as the urban street children and the manual scavengers. The report also discusses India's position in the Global Hunger Index and the state of malnutrition in the country.

10.7 TIMES-SERIES DATA

Time Series data, as the name suggests, is a sequence of data arranged or listed in order of their time. It is usually collected over a very long time and presented in the form of lists, tables or graphs which can be measured and subjected to meaningful analysis of trends, patterns and extrapolation of data to make predictions about the future. For instance, if data is available of, say, 100 years of rainfall or weather, one can find out cycles of extreme patterns and even predict about the future. Time series is used in economics, weather, rainfall, census, earthquakes

etc. Many data journalists with mathematics, statistics or econometrics backgrounds can make sense of data more easily but a refresher course can familiarise almost all students in understanding and analysing long-term data. This allows students to develop data analysis skills as part of a framework of critical thinking.

10.7.1 Following Exercises may Prepare Students to Look at the Time-Series Data Critically:

1. Watch the TED video link below of the famous data scientist, physician and statistician Hans Rosling and write 500 words on the co-relations between population and prosperity around the world along with supporting data.
2. https://www.ted.com/talks/hans_rosling_shows_the_best_stats_you_ve_ever_seen?language=en
3. Go to the website of the Census of India and spot trends and patterns about the co-relations between population and prosperity in India on the above lines
4. Visit the website of the National Sample Survey Organisation (NSSO) which conducts large-scale sample surveys in diverse fields and write a note on the range of subjects covered.
5. See the latest Annual Report of the Reserve Bank of India (on the RBI website) and prepare a note on the state of the economy and what it says about the data on investment climate and growth in the past year
6. Go through the latest Economic Survey and list five most critical areas of growth in the Indian economy in the last year
7. List five most important types of data collected by the National Family Health Survey (NFHS) and write a short note on its methods of data collection
8. Go through the website of the Bureau of Police Research and Development (BPRD) and the National Crime Records Bureau (NCRB) and make a list of five most important types of data collected about crime and prisons in India.
9. Go through the exit polls conducted by different agencies and media organisations during the last Assembly elections in India and compare them for accuracy.

Check Your Progress: 2

Note: 1) Use the space given below for your answer

2) Compare your answer with those given at the end of this unit.

1. Explain how does data regulation can help in protecting the citizens' rights?

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10.8 SIMPLIFYING THE CHALLENGES OF DATA JOURNALISM

One of the lessons of data journalism is that it is skill oriented work which can be learnt by following a method. It is never easy to deal with numbers when they come in huge quantities or in the form of time-series. However, it is no rocket science either; even the most accomplished researchers and data scientists ensure data integrity, and check and recheck data before coming to conclusions. What must be always kept in mind is that data is not something unusual but it is a part of everyday life. It is impossible to understand government policies without an understanding of data. It is also difficult to nail the truth about environmental issues like polluting factories of climate change or medical data like cancer research and spread of diseases without deploying long-term data. Fortunately, there are enough web-based resources available to modern journalism students to hone their skills. It is a huge relief that once basic proficiency is acquired in making sense of data, the process can be very enlightening and empowering. No politician or bureaucrat can deceive or delude a data journalist by bombarding her with data.

10.8.1 Some Important Indian Websites/ Outlets Useful For Data Journalism

- a) India Spend <http://www.indiaspend.com/>
- b) Inclusive Media for Change <http://www.im4change.org/>
- c) Peoples' Archive of Rural India (PARI) <https://ruralindiaonline.org/>
- d) South Asia Terrorism Portal <http://www.satp.org/>
- e) Centre for Science and Environment <https://www.cseindia.org/>
- f) E-Social Sciences (online scholarly resource) <http://www.esocialsciences.org/Home/NewIndex.aspx>

10.8.2 Important International web Resources on Data Journalism

Resources for Learning and Doing Data Journalism (Press Institute of America database) <https://www.americanpressinstitute.org/publications/reports/strategy-studies/data-resources/>

“Knowing the Numbers,” (Harvard’s Journalist’s Resource project) <http://isoj.org/research/knowing-the-numbers-assessing-attitudes-among-journalists-and-educators-about-using-and-interpreting-data-statistics-and-research/>

Video: “Solve Every Statistics Problem with One Weird Trick,” NICAR 2016, Jonathan Stray <https://www.youtube.com/watch?v=3TIJ6KVImF0>

The Upshot (The New York Times) <https://www.nytimes.com/section/upshot>
Stanford Computational Journalism Lab <http://cjlaboratory.stanford.edu/>

Data Is Beautiful, a community on Reddit
<https://www.reddit.com/r/dataisbeautiful/>

Data is Plural. Sign up for <http://tinyletter.com/data-is-plural>. All datasets can be found in an updated master spreadsheet.

Jonathan Stray, The Curious Journalist's Guide to Data, 2016.
<https://towcenter.org/research/the-curious-journalists-guide-to-data/> Brant Houston, Computer-Assisted Reporting: A Practical Guide, 2014.

David Herzog, Data Literacy: A User's Guide, 2016. <https://study.sagepub.com/herzog>

ProPublica <https://www.propublica.org/>

Video: "The Newest Muckrakers: Investigative Reporting in the Age of Data Science," Sarah Cohen, C+J Symposium Stanford, 2016

<http://journalism.stanford.edu/cj2016/#livestream>

10.9 LET US SUM UP

This Unit has covered certain core skills and a broader understanding of data, an analysis of its sociology and politics, and a range of benefits arising out of its presentation and interpretation. A precondition to practicing data journalism is to understand that the art and science of data journalism requires proficiency in specific skills, right from simple articulation of numbers to an understanding of spreadsheet programmes to its visualisation and interpretation for meaningful analyses. The same has been simplified and conveyed to you through this unit.

10.10 KEY WORDS

Data Journalism : Application of data science to governance.

Data Science : The study of extraction of knowledge from data.
News Aggregators/

Feed Aggregators : Mass piece of information the form of text, audio and video clips and still pictures presented to existing and potential users to indifferent formats to save time and effort.

10.11 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

Refer to 10.2.1 and 10.2.2

Check Your Progress: 2

Refer to 10.4.1